THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF ENGINEERING & TECHNOLOGY
(RSET)

DEPARTMENT OF CIVIL ENGINEERING

COURSE STRUCTURE & SYLLABUS

FOR

Bachelor of Technology
in
Civil Engineering

W.E.F

AY - 2026 – 30

THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

Department of Civil Engineering
Royal School of Engineering and Technology
The Assam Royal global University

Course Structure

Semester I						
Sl No	**Course Title**	**Course Code**	**L**	**T**	**P**	**Credits**
1	Chemistry	CHY022C101	3	0	0	3
2	Chemistry Lab	CHY022C111	0	0	2	1
3	Mathematics - I	MAT022C101	4	0	0	4
4	Fundamentals of Programming	CEE022C103	3	0	0	3
5	Fundamentals of Programming Lab	CEE022C113	0	0	2	1
6	Engineering Graphics & Design Lab	CEE022C114	0	0	4	2
7	Manufacturing Workshop Practice	MEE022C115	0	0	4	2
8	Universal Human Values	BHS022A106	2	0	0	**2**
9	Sports and Yoga /NCC/NSS	CEE022S117	0	0	2	1
						19
Semester II						
Sl No	**Course Title**	**Course Code**	**L**	**T**	**P**	**Credits**
1	Physics	PHY022C201	4	0	0	4
2	Physics Lab	PHY022C211	0	0	2	1
3	Mathematics - II	MAT022C202	4	0	0	4
4	Basic Electrical Engineering	CSE022C205	3	0	0	3
5	Basic Electrical Engineering Lab	CSE022C215	0	0	2	1
6	Engineering Mechanics	CEE022C217	4	0	0	4
7	English for Technical Writing	CEN982A203	2	0	0	2
8	Design Thinking & Ideation Lab	COD022S216	0	0	2	1

						20
Semester III						
Sl No	**Course Title**	**Course Code**	**L**	**T**	**P**	**Credits**
1	Solid Mechanics	CEE022C301	3	0	0	3
2	Strength of Materials Lab	CEE022C311	0	0	2	1
3	Building Planning & CAD	CEE022C303	3	0	0	3
4	Fluid Mechanics	CEE022C304	3	0	0	3
5	Fluid Mechanics Lab	CEE022C314	0	0	2	1
6	Concrete Technology	CEE022C305	3	0	0	3
7	Concrete Technology Lab	CEE022C315	0	0	2	1
8	Mathematics- III	MAT022C306	3	0	0	3
9	IKS	IKS022C305	2	0	0	2
						20
Semester IV						
Sl No	**Course Title**	**Course Code**	**L**	**T**	**P**	**Credits**
1	Structural Analysis I	CEE022C401	4	0	0	4
2	Hydraulic Engineering	CEE022C402	3	0	0	3
3	Hydraulic Engineering Lab	CEE022C412	0	0	2	1
4	Transportation Engineering	CEE022C403	3	0	0	3
5	Transportation Engineering Lab	CEE022C413	0	0	2	1
6	Construction Engineering & Management	CEE022C405	3	0	0	3
7	Geotechnical Engineering	CEE022C406	3	0	0	3
8	Geotechnical Engineering Lab	CEE022C416	0	0	2	1
9	Surveying and Geomatics	CEE022C404	3	0	0	3
10	Surveying and Geomatics Lab	CEE022C414	0	0	2	1
						23
Semester V						
Sl No	**Course Title**	**Course Code**	**L**	**T**	**P**	**Credits**
1	Design of RCC Structures	CEE022C501	4	0	0	4
2	Environmental Engineering	CEE022C502	3	0	0	3
3	Environmental Engineering Lab	CEE022C512	0	0	2	1
4	Engineering Economics, Estimation & Costing	CEE022C503	3	0	0	3
5	Hydrology & Water Resource Engineering	CEE022C504	3	0	0	3
6	Hydrology Lab	CEE022C513	0	0	2	1

7	Program elective I	CEE022C505	3	0	0	3
8	Open Elective I	CEE022M505	3	0	0	3
						21
Semester VI						
Sl No	**Course Title**	**Course Code**	**L**	**T**	**P**	**Credits**
1	Design of Steel Structures	CEE022C601	4	0	0	4
2	Sustainable & Green Construction	CEE022C603	3	0	0	3
3	Structural Analysis II	CEE022C604	4	0	0	4
4	Program Elective- 2 (Basket)	CEE022D60X	3	0	0	3
5	Program Elective- 3 (Basket)	CEE022D60X	3	0	0	3
6	Program Elective- 4 (Basket)	CEE022D60X	3	0	0	3
7	Open Elective II	XX(OEC)	3	0	0	3
						23
Semester VII						
Sl. No.	**Course Title**	**Course Code**	**L**	**T**	**P**	**Credits**
1	Foundation Engineering	CEE022C70X	4	0	0	4
2	Program Elective-5(Basket)	CEE022D60X	3	0	0	3
3	Program Elective- 6 (Basket)	CEE022D60X	3	0	0	3
4	Open Elective III	CEE022070X	3	0	0	3
5	Minor Project	CEE022C721	0	0	8	4
6	INTERNSHIP	CEE022C801	0	0	12	4
						21
Semester VIII						
Sl No	**Course Code**	**Course Title**	**L**	**T**	**P**	**Credits**
1	Program Elective- 7 (Basket)	CEE022D60X	3	0	0	3
2	Program Elective- 8 (Basket)	CEE022D60X	3	0	0	3
3	Major Project	CEE022C802	0	0	20	16
						22
Total						**169**

Program Elective Courses

Elective No	Sl. No	Subject Code	Name of the Elective
I	1		Pavements Materials
	2		Smart Cities
	3		Open Channel Flow
II	1		Reliability Analysis of Structures
	2		Biological Processes for Contaminant Removal
	3		Sustainable Construction Methods
III	1		Energy Efficient Buildings
	2		Traffic Engineering & Management
	3		Remote Sensing
IV	1		Railway Engineering
	2		Ground Improvement Techniques
	3		Airport Planning & Design
V	1		Industrial Waste Management
	2		Structural Dynamics
	3		Earth Retaining Structures
VI	1		Reliability Analysis of Structures
	2		Irrigation Engineering
	3		Earthquake Engineering
VII	1		Robotics and Automation for Civil Engineers
	2		Satellite Image Processing
	3		Restoration and Rehabilitation of Structures
VIII	1		Prestressed Concrete
	2		Construction Project Planning & System
	3		Energy Efficient Buildings

Open Elective Courses

Sl. No	Subject Code	Name of the Open Elective (New curriculum)
1		Steel concrete composite structures
2		Finance for engineering
3		Earth and rockfill dams
4		Computational fluid dynamics
5		Rainwater harvesting
6		Transport and environment
7		Environment Quality monitoring
8		Evaluating accessibility/universal design in built environments
9		Disaster risk Reduction and Management

SEMESTER -I

DETAILED SYLLABUS OF 1st SEMESTER

Subject Name: Chemistry	**Subject Code: CHY022C101**
L-T-P-C – 3-0-0-3	**Credit Units: 03 Scheme of Evaluation: T**

Objective:

The objectives of the course are to s to acquaint the students with the basic phenomenon/concepts of chemistry, the student faces during course of their study in the industry and Engineering field and to understand the new developments and breakthroughs efficiently in engineering and technology.

Prerequisites: Concepts of +2 level Chemistry

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic properties of chemical reactions	**BT 1**
CO 2	**Interpret** periodic properties such as ionization potential, electronegativity, oxidation states, electronegativity and bulk properties and processes using thermodynamic considerations	**BT 2**
CO 3	**Experiment with** major chemical reactions that are used in the synthesis of molecules.	**BT 3**
CO 4	**Analyze** microscopic chemistry in terms of atomic and molecular orbitals and intermolecular Forces	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course Content**	**Periods**
I.	**Atomic and Molecular Structure**	Schrodinger equation. Particle in a box solution and their applications for conjugated molecules and nanoparticles. Forms of the hydrogen atom wave functions and the plots of these functions to explore their spatial variations. Molecular orbitals of diatomic molecules and plots of the multicenter orbitals. Equations for atomic and molecular orbitals. Energy level	**16**
II.		Principles of spectroscopy and selection rules. Electronic spectroscopy. Fluorescence and its applications in	**17**

	Spectroscopic Techniques and Applications, Intermolecular Forces and Potential Energy Surfaces	medicine. Vibrational and rotational spectroscopy of diatomic molecules. Applications. Nuclear magnetic resonance and magnetic resonance imaging, surface characterization techniques. Diffraction and scattering. Ionic, dipolar and van Der Waals interactions. Equations of state of real gases and critical phenomena. Potential energy surfaces of H3, H2F and HCN and trajectories on these surfaces.	
III.	**Use of free Energy in Chemical Equilibria and Periodic Properties**	Thermodynamic functions: energy, entropy and free energy. Estimations of entropy and free energies. Free energy and EMF. Cell potentials, the Nernst equation and applications. Acid base, oxidation reduction and solubility equilibria. Water chemistry. Corrosion. Use of free energy considerations in metallurgy through Ellingham diagrams. Effective nuclear charge, penetration of orbitals, variations of s, p, d and f orbital energies of atoms in the periodic table, electronic configurations, atomic and ionic sizes, ionization energies, electron affinity and electronegativity, polarizability, oxidation states, coordination numbers and geometries, hard soft acids and bases, molecular geometries	**16**
IV.	**Stereochemistry, Organic Reactions and Synthesis of a Drug Molecule**	Representations of 3 dimensional structures, structural isomers and stereoisomers, configurations and symmetry and chirality, enantiomers, diastereomers, optical activity, absolute configurations and conformational analysis. Isomerism in transitional metal compounds Introduction to reactions involving substitution, addition, elimination, oxidation, reduction, cyclization and ring openings. Synthesis of a commonly used drug molecule	**17**
TOTAL			**66**

Subject Name: Chemistry lab		Subject Code: CHY022C111
L-T-P-C – 0-0-2-1	Credit Units: 01	Scheme of Evaluation: P

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Determination of surface tension and viscosity.
2. Thin layer chromatography.
3. Ion exchange column for removal of hardness of water.
4. Determination of chloride content of water.
5. Colligative properties using freezing point depression.
6. Determination of the rate constant of a reaction.
7. Determination of cell constant and conductance of solutions.
8. Potentiometry - determination of redox potentials and EMFs.
9. Synthesis of a polymer/drug.
10. Saponification/acid value of an oil.
11. Chemical analysis of a salt.
12. Lattice structures and packing of spheres.
13. Models of potential energy surfaces.
14. Chemical oscillations- Iodine clock reaction.
15. Determination of the partition coefficient of a substance between two immiscible liquids.
16. Adsorption of acetic acid by charcoal.
17. Use of the capillary viscosimeters to the demonstrate of the isoelectric point as the pH of minimum viscosity for gelatin sols and/or coagulation of the white part of egg.

Text Books

1. *A Textbook of Physical Chemistry*, Negi A.S. and Anand S.C., 2nd *Edition,* 2007, New Age International
2. *Concise Inorganic Chemistry,* Lee J.D., 5th Edition, 2008, John Wiley and Sons Ltd.

Reference Books:

1. Atkins, P.W. and Paula, J. De,*Physical Chemistry,* 10th Edition, 2014, Oxford University Press
2. Huheey, J.E. Keiter, E.A. Keiter, R.L Medhi, O.K.,*Inorganic Chemistry Principles of Structure and Reactivity*, 4th Edition, 2006, Pearson Education

Subject Name: Mathematics-I **Subject Code: MAT022C101**
L-T-P-C – 4-0-0-4 **Credit Units: 04** **Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach the students Mathematics fundamentals necessary to formulate, solve and analyze engineering problems.

Prerequisites: Concepts of Mathematics of +2 level

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** basic terminologies of calculus	**BT 1**
CO 2	**Understand** the applications of differential and integral calculus in different fields of Engineering.	**BT 2**
CO 3	**Apply** the single and multivariable differential and Integral calculus in engineering problems.	**BT 3**
CO 4	**Analyze** and **assess** the patterns in series.	**BT 4 & 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I	**Basic Calculus:**	Curvature, evolutes and involutes; Evaluation of definite and improper integrals; Beta and Gamma functions and their properties; Applications of definite integrals to evaluate surface areas and volumes of revolutions. Rolle's Theorem, Mean value theorems and applications; Extreme values of functions; Linear approximation; Indeterminate forms and L'Hospital's rule.	**10**
II	**Sequences and Series:**	Limits of sequence of numbers, Calculation of limits, Infinite series; Tests for convergence; Power series, Taylor and Maclaurin series; Taylor theorem, convergence of Taylor series, error estimates.	**20**
III	**Multivariable Calculus (Differentiation):**	Limit, continuity and partial derivatives, directional derivatives, gradient, total derivative; Tangent plane and normal line; Maxima, minima and saddle points; Method of Lagrange multipliers.	**10**
IV	**Multivariable Calculus (Integration):**	Multiple Integration: Double integrals (Cartesian), change of order of integration in double integrals, Change of variables (Cartesian to polar), Applications: areas and volumes, Center of mass and Gravity (constant and variable densities); Triple integrals (Cartesian), orthogonal curvilinear coordinates, Simple applications involving cubes, sphere and rectangular parallelepipeds;	**20**

		Scalar line integrals, vector line integrals, scalar surface integrals, vector surface integrals, Gradient, curl and divergence, Theorems of Green, Gauss and Stokes.	
TOTAL			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 15 NCH = 60 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. Reena Garg, Engineering Mathematics, Khanna Book Publishing Company, 2022.
2. Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021.
3. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
4. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
5. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
6. Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
7. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
8. A text book of Engineering Mathematics, Bali N. P. and Narayan Iyenger N., 9th Edition, 2016, Laxmi Publication.
9. Mathematical Methods for Physics and Engineering: A Comprehensive Guide, K. F. Riley, M. P. Hobson, 3rd Edition, 2006, Cambridge University Press

Reference Books:

1.Grewal B. S., *Higher Engineering Mathematics,* 43rd Edition, 2014, Khanna Publishers.
2.Raisinghania M.D., *Ordinary and Partial Differential Equations*, 17th Edition, 2014, S. Chand and Co., New Delhi.
3.Narayna S., *A Text Book of Vector Calculus,* Revised Edition, 2009, S. Chand & Co., New Delhi.

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/R-18%20Mathematics-II.pdf
2. http://www.bosecuttack.in/studentcorner/LECTURE_NOTE.MATH2.2ND_SEM__1_.pdf
3. https://www.srividyaengg.ac.in/coursematerial/Iyear/111223.pdf

Subject: Fundamentals of Programming	**Subject Code: CSE022C104**	
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students capable of using C programming to solve basic as well as advanced computing problems.

Prerequisites: None

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**List** the various constructions used in programming	**BT 1**
CO 2	**Demonstrate** the working of C programming language.	**BT 2**
CO 3	**Apply** the programming concepts to solve various problems.	**BT 3**
CO 4	**Analyze** and debug the errors while writing the programs.	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Hours**
I	**Fundamentals of Programming**	Introduction to Programming; Introduction to components of a computer system (disks, memory, processor, where a program is stored and executed, operating system, compilers etc.) Idea of Algorithm: steps to solve logical and numerical problems. Representation of Algorithm: Flowchart/Pseudocode with examples. From algorithms to programs; source code, variables (with data types) variables and memory locations, Syntax and Logical Errors in compilation, object and executable code.	**15**
II	**Expressions, Conditional Operators and Loops**	Arithmetic expressions and precedence. Conditional Branching and Loops. Writing and evaluation of conditionals and consequent branching. Iteration and loops. Arrays, Arrays (1-D, 2-D), Character arrays and Strings	**15**
III	**Functions, Recursion, Sorting**	Basic Algorithms, Searching, Basic Sorting Algorithms (Bubble, Insertion and Selection), Finding roots of equations, notion of order of complexity through example programs (no formal definition required). Function, Functions (including using built in libraries), Parameter passing in functions, call by value, Passing arrays to functions: idea of call by reference. Recursion, Recursion as a different	**15**

		way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc. Quick sort or Merge sort.	
IV	**Advanced Programming Concepts using C**	Structures, Defining structures and Array of Structures, Pointers, Idea of pointers, Defining pointers, Use of Pointers in self-referential structures, notion of linked list (no implementation), File handling.	**15**
TOTAL			**60**

Subject: Fundamentals of Programming lab		**Subject Code: CSE022C114**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Programming for Problem Solving Lab Syllabus CSE022C114

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Character set, Tokens, Keywords and Identifiers, Constants, variables, data types, statements, comments, declaration of storage class, assigning values to variables.
2. Managing I/O, reading and writing characters, formatted Input/output.
3. Arithmetic operators, relational operators, logical operators, assignment operators, increment & decrement operators, conditional operators, bitwise operators, special operators.
4. Importance of decision making, decision making with if statement, if-else statement, nested if-else statements, switch-case statement.
5. Importance of iterative statements, the while statement, do-while statement, for statement, nested for looping.
6. Significance of Arrays, creation and use of one & two-dimensional arrays
7. Declaration and use of string variables, reading and writing strings.
8. Benefits of user-defined functions, creation and use of user-defined functions, parameter passing, return types.
9. Use of Pointers, declaration & initialization of pointer variables, accessing a variable through its pointer.
10. Defining, opening & closing files in C.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	2*15 NCH = 30 NCH	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Computer Fundamentals and Programming in C,* Reema Thareja, 2nd Edition, 2016, Oxford University Press, Delhi.

Reference Books:

1. E Balaguruswamy, *Computing Fundamentals and C Programming*, 1st Edition, 2017, McGraw Hill.
2. Venugopal and Prasad, *Mastering C*, 2nd Edition, 2017, Tata McGraw Hill.
3. Yashawant Kanetkar,*Let us C,* 15th Edition, 2017, BPB.

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Programming%20for%20Problem%20Solving.pdf
2. NPTEL course on Introduction to Programming in C by Prof. Satyadev Nandakumar, IIT, Kanpur
3. NPTEL course on Problem Solving Through Programming in C by Prof. Anupam Basu, IIT, Kharagpur

Subject Name: Engineering Graphics & Design Lab	Subject Code: CEE022C114
L-T-P-C – 0-0-4-2	Credit Units: 04 Scheme of Evaluation: P

Objective:

The objectives of the course are to make students understand the process of drawing projections and sections and basic engineering drawing formats and to convert sketches to engineered drawings.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Relate** with the concepts of drawings and projections	**BT 1**
CO 2	**Understand** the dimension and figures using the drawing instruments and acquire visualisation skills, projection of	**BT 2**
CO 3	**Utilize** engineering curves in tracing the paths of simple machine components.	**BT 3**
CO 4	**Analyse** and assess sketches to convert them to engineered drawings.	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 60 (4 hours per week)

Minimum 10 Laboratory experiments based on the following-

Objective: To determine and verify different parameter of Fluid

Experiment	No.
To prepare a sheet on Lettering, dimensions & Scales	01
To prepare different types scales used in Engineering Drawings	02
To prepare curves used in Engineering Graphics	03
To prepare a sheet on Projection of point & lines	04
To prepare a sheet on Projection of planes	05
To prepare a sheet on Projection of solids	06
To prepare a sheet on orthographic & Isometric projections	07
To draw basic geometrical entities using Autocad software	08

To draw Orthographic projections in Autocad Software	09
To draw Isometric projections in Autocad Software	10

Text Books:

1. Engineering Drawing; Bhatt, N.D, 53rd Edition, 2016, Charotar Publishing House

Reference Books:

1. Jolhe Dhananjay A; Engineering drawing, 5th Edition, 2010, Tata McGraw-Hill Education Pvt. Ltd., New Delhi

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Engineering%20Graphics%20Manual%20final.pdf
2. https://www.pvpsiddhartha.ac.in/autonomus14/1-1/it/IT1L3.pdf
3. NPTEL Course on Engineering Drawing and Computer Graphics by Prof. Rajaram Lakkaraju, IIT, Kharagpur
4. NPTEL Course on Engineering Graphics by Prof. Nihar Ranjan Patra, IIT, Kanpur

Subject Name: Manufacturing Practices Workshop Subject Code: MEE022C115

L-T-P-C – 0-0-4-2 Credit Units: 02 Scheme of Evaluation: P

Objective:

The objectives of the course are to provide exposure to the students with hands-on experience on various basic engineering practices in Civil, Mechanical, Electrical and Electronics Engineering.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Label** the various techniques used under mechanical engineering	**BT 1**
CO 2	**Understand** the different manufacturing processes which are commonly employed in the industry	**BT 2**
CO 3	**Utilize** tools, instruments and techniques learnt to perform basic household chores in terms of house wiring, carpentry etc	**BT 3**
CO 4	**Experiment** using the tools and techniques learnt for various purposes and decide on the best prospect.	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 40 (4 hours per week)

Minimum 10 Laboratory experiments based on the following-

The lecture sessions will be on the following topics:

- Manufacturing Methods- casting, forming, machining, joining, advanced manufacturing methods.
- CNC machining, Additive manufacturing.
- Fitting operations & power tools.
- Electrical & Electronics.
- Carpentry.
- Plastic moulding, glass cutting.
- Metal casting.
- Welding (arc welding & gas welding), brazing g topics:

And the lab sessions will on the topics:

- Machine shop
- Fitting shop
- Carpentry
- Electrical & Electronics
- Welding shop (Arc welding + Gas welding)
- Casting
- Smithy
- Plastic moulding & Glass Cutting

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
-	4*15 NCH = 60 NCH	20 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. Elements of Workshop Technology, Hajra Choudhury, S K, Hajra Choudhury, A K, 14th Edition, 2007, Mumbai Media Promoters
2. Manufacturing Technology – I, Gowri P. Hariharan and A. Suresh Babu, 2008, Pearson Education.

Reference Books:

1. Roy A. Lindberg, Processes and Materials of Manufacture", 4th Edition, 1998, Prentice Hall India,

Additional Readings:

1. http://mm-coep.vlabs.ac.in/LaserSpotWelding/Theory.html?domain=Mechanical%20Engineering&lab=Welcome%20to%20Micromachining%20laboratory
2. http://fab-coep.vlabs.ac.in/exp7/Theory.html?domain=Mechanical%20Engineering&lab=Welcome%20to%20FAB%20laboratory

Subject Name: Universal Human Values		Subject Code: BHS022A106
L-T-P-C – 2-0-0-2	Credit Units: 02	Scheme of Evaluation: T

Objective:

The objectives of the course are to help the students appreciate the essential complementarily between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic need of human values in real life	**BT 1**
CO 2	**Understand** the importance of following the basic universal human values	**BT 2**
CO 3	**Apply** the holistic understanding in one's day-today life so as to keep oneself happy and to socialize with nature, society, etc	**BT 3**
CO 4	**Analyze** the harmony within human beings by distinguishing the needs of the self and the body.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Value Education**	Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education), Understanding Value Education, Sharing about Oneself, Self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human Aspirations, Exploring Human Consciousness, Happiness and Prosperity – Current Scenario, Lectured, Method to fulfil the Basic Human Aspirations, Exploring Natural Acceptance	**11**
II	**Harmony in Human Being**	Understanding Human being as the Co-existence of the Self and the Body, distinguishing between the Needs of the Self and the Body, Exploring the difference of Needs of Self and Body, The Body as an Instrument of the Self Understanding Harmony	**11**

		in the Self, Exploring Sources of Imagination in the Self, Harmony of the Self with the Body, Programme to ensure self-regulation and Health, Exploring Harmony of Self with the Body	
III	**Harmony in the Family & Society**	Harmony in the Family – the Basic Unit of Human Interaction, 'Trust' – the Foundational Value in Relationship", Exploring the Feeling of Trust, 'Respect' – as the Right Evaluation, Exploring the Feeling of Respect, Other Feelings, Justice in Human-to-Human Relationship Understanding Harmony in the Society, Vision for the Universal Human Order, Exploring Systems to fulfil Human Goal	**11**
IV	**Harmony in Nature & Implications of Holistic Understanding**	Understanding Harmony in the Nature, Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature, Exploring the Four Orders of Nature, Realizing Existence as Co-existence at All Levels, The Holistic Perception of Harmony in Existence, Exploring Co-existence in Existence. Natural Acceptance of Human Values, Definitiveness of (Ethical) Human Conduct, Exploring Ethical Human Conduct, A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order, Competence in Professional Ethics, Exploring Humanistic Models in Education, Holistic Technologies, Production Systems and Management Models-Typical Case Studies, Strategies for Transition towards Value-based Life and Profession Exploring Steps of Transition towards Universal Human Order	**11**
TOTAL			**44**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2 * 22 NCH = 44 NCH	-	8 * 2 NCH = 16 NCH (Seminar, Case Study, Discussion, Internship)

Text Books:

1. *A Foundation Course in Human Values and Professional Ethics,* R R Gaur, R Asthana, G P Bagaria, 2nd Revised Edition, Excel Books, New Delhi

Reference Books:

1. *Human Values*, A.N. Tripathi, 3rd Edition, 2019, New Age Intl. Publishers, New Delhi,

Additional Readings:

1. https://uhv.org.in/uhv2notes
2. https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php

Subject Name: Sports and Yoga		Subject Code: CEE022C117
L-T-P-C – 0-0-2-1	Credit Units: 01	Scheme of Evaluation: P

Objective:

The objectives of the course are to make the students understand the importance of sound health and fitness principles as they relate to better health, to expose the students to a variety of physical and yogic activities aimed at stimulating their continued inquiry about Yoga, physical education, health and fitness and to develop among students an appreciation of physical activity as a lifetime pursuit and a means to better health.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Choose** the best form of yoga/ exercise for them	**BT 1**
CO 2	**Understand** basic skills associated with yoga and physical activities including strength and flexibility, balance and coordination	**BT 2**
CO 3	**Experiment with** different forms of yoga to keep oneself physically fit and mentally strong	**BT 3**
CO 4	**Assess** current personal fitness levels	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Physical Education, Olympic Movement, Fitness, Wellness & Lifestyle**	Meaning & definition of Physical Education. Aims & Objectives of Physical Education. Changing trends in Physical Education, Ancient & Modern Olympics (Summer & Winter), Olympic Symbols, Ideals, Objectives & Values, Awards and Honours in the field of Sports in India (Dronacharya Award, Arjuna Award, Dhayanchand Award, Rajiv Gandhi Khel Ratna Award etc., Meaning & Importance of Physical Fitness & Wellness, Components of Physical fitness, o Components of Health-related fitness, Components of wellness, Preventing Health Threats through Lifestyle Change, Concept of Positive Lifestyle.	**5**
II	**Anatomy & Physiology in Physical Education, Sports,**	Define Anatomy, Physiology & Its Importance, ffect of exercise on the functioning of Various Body Systems. (Circulatory System, Respiratory System, Neuro-Muscular System etc.), Meaning & Importance of Kinesiology & Biomechanics in Physical Edu. & Sports,	**5**

	Yoga & Postures	o Newton's Law of Motion & its application in sports. o Friction and its effects in Sports, Meaning and Concept of Postures, Causes of Bad Posture. Advantages & disadvantages of weight training. Concept & advantages of Correct Posture. Common Postural Deformities – Knock Knee; Flat Foot; Round Shoulders; Lordosis, Kyphosis, Bow Legs and Scoliosis. Corrective Measures for Postural Deformities	
III	**Yoga & Lifestyle**	Meaning & Importance of Yoga, Elements of Yoga, Asanas, Pranayama, Meditation & Yogic Kriyas, yoga for concentration & related Asanas (Sukhasana; Tadasana; Padmasana & Shashankasana) Relaxation Techniques for improving concentration - Yog-nidra, Asanas as preventive measures. Hypertension: Tadasana, Vajrasana, Pavan Muktasana, Ardha Chakrasana, Bhujangasana, Sharasana. Obesity: Procedure, Benefits & contraindications for Vajrasana, Hastasana, Trikonasana, Ardh Matsyendrasana. Back Pain: Tadasana, Ardh Matsyendrasana, Vakrasana, Shalabhasana, Bhujangasana. Diabetes: Procedure, Benefits & contraindications for Bhujangasana, Paschimottasana, Pavan Muktasana, Ardh Matsyendrasana. Asthema: Procedure, Benefits & contraindications for Sukhasana, Chakrasana, Gomukhasana, Parvatasana, Bhujangasana, Paschimottasana, Matsyasana.	**5**
IV	**Training, Planning and Psychology in Sports**	Meaning of Training, Warming up and limbering down, Skill, Technique & Style, Meaning and Objectives of Planning. Tournament – Knock-Out, League/Round Robin & Combination. Definition & Importance of Psychology in Physical Edu. & Sports, Define & Differentiate Between Growth & Development, Adolescent Problems & Their Management, Emotion: Concept, Type & Controlling of emotions, Meaning, Concept & Types of Aggressions in Sports. Psychological benefits of exercise. Anxiety & Fear and its effects on Sports Performance. Motivation, its type & techniques. Understanding Stress & Coping Strategies	**5**
TOTAL			**20**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
	1*20 NCH = 20 NCH	10 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. *Modern Trends and Physical Education*, Ajmer Singh, Gill J.S, Bains J, 4th Edition, 2012, Kalyani Publishers

Reference Books:

1. B.K.S. Iyengar*, Light on Yoga*, 2006, Thorsons

SEMESTER -II

DETAILED SYLLABUS OF 2nd SEMESTER

Subject Name: Physics		**Subject Code: PHY022C201**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students enhance the fundamental knowledge in Physics and its applications relevant to various streams of Engineering and Technology

Prerequisites: Concepts of Physics of +2 level

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Study** the basic concepts of Physics.	**BT 1**
CO 2	**Understand** the basic concepts of Physics.	**BT 2**
CO 3	**Solve** problems in wave mechanics.	**BT 3**
CO 4	**Analyse** knowledge in calculating red and blue shift and also in acoustics.	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I.	**Classical Mechanics and Dynamics**	Transformation of scalars and vectors under Rotation transformation; Forces in Nature; Newton's laws and its completeness in describing particle motion; Form invariance of Newton's Second Law; Solving Newton's equations of motion in polar coordinates; Problems including constraints and friction; Extension to cylindrical and spherical coordinates. Potential energy function; F = - Grad V, equipotential surfaces and meaning of gradient; Conservative and non- conservative forces, curl of a force field; Central forces; Conservation of Angular Momentum; Energy equation and energy diagrams; Elliptical, parabolic and hyperbolic orbits; Kepler problem; Application: Satellite manoeuvres.	**17**
II.	**Advanced Dynamics and Oscillatory Motion**	Non-inertial frames of reference; Rotating coordinate system: Five-term acceleration formula. Centripetal and Coriolis accelerations; Applications: Weather systems, Foucault pendulum; Harmonic oscillator; Damped harmonic motion – over-damped, critically damped and	**16**

		lightly- damped oscillators; Forced oscillations and resonance.	
III.	**Rigid Body Dynamics and Kinematics**	Definition and motion of a rigid body in the plane; Rotation in the plane; Kinematics in a coordinate system rotating and translating in the plane; Angular momentum about a point of a rigid body in planar motion; Euler's laws of motion, their independence from Newton's laws, and their necessity in describing rigid body motion; Examples.	**16**
IV	**Advanced Rigid Body Dynamics: Three-Dimensional Motion**	Introduction to three-dimensional rigid body motion — only need to highlight the distinction from two-dimensional motion in terms of (a) Angular velocity vector, and its rate of change and (b) Moment of inertia tensor; Three-dimensional motion of a rigid body wherein all points move in a coplanar manner: e.g. Rod executing conical motion with center of mass fixed — only need to show that this motion looks two-dimensional but is three-dimensional, and two-dimensional formulation fails.	**17**
TOTAL			**66**

Physics Lab Syllabus

Subject Name: Physics Lab	**Subject Code: PHY022C211**	
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Detailed Syllabus:

Experiment	Experiment Title	Lab Hours
I	Determination of Moment of Inertia of a given solid about its own axis by using M.I.Table	2
II	Determination of Young's Modulus using Searle's Apparatus	2
III	Determination of Rigidity of Modulus of the material of the given rod by Statistical method	2
IV	Determination of Powers of Given lenses using an Optical Bench i. Concave Lens, ii Convex Lens	2
V	Determination of Resistance of a Galvanometer using Post Office Box.	2
VI	To determine the mechanical equivalent of heat by Joule's calorimeter	2

VII	Determination of ratio of E.M.F of two cells using Potentiometer.	2
VIII	To determination of the focal length of a convex mirror with the help of an auxiliary lens.	2
IX	Determination of Horizontal Components of Earth's Magnetic field using Magnetometer	2
X	Determination of coefficient of Viscosity of water by Capillary Flow Method	2

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. Engineering Mechanics, 2nd ed. – D.S. Bedi, M.P. Poonia
2. Basic Mechanical Engineering – S.C. Sharma, M.P. Poonia
3. Engineering Mechanics, 2nd ed. — MK Harbola
4. Introduction to Mechanics — MK Verma
5. An Introduction to Mechanics — D Kleppner & R Kolenkow
6. Principles of Mechanics — JL Synge & BA Griffiths
7. Mechanics — JP Den Hartog
8. Engineering Mechanics - Dynamics, 7th ed. - JL Meriam
9. Mechanical Vibrations — JP Den Hartog
10. Theory of Vibrations with Applications — WT Thomson

Reference Books:

1. Singh A.K. and Malik Hitendra Engineering Physics, 2nd Edition, 2016, McGraw Hill Education Private Limited. New Delhi.
2. Gaur R.K and Gupta S.L, Engineering Physics, 2015, Dhanpat Rai publication, New Delhi.
3. Arthur Beiser, Shobhit Mahajan, S. Rai. Choudhury, Concept of Modern physics, 6th Edition, 2009, McGraw-Hill education Private limited. New Delhi.
4. M Ghosh & D Bhattacharya, A Textbook of Oscillations, Waves and Acoustics, 5th Edition, 2016, S. Chand publication.

Additional Readings

1. https://www.griet.ac.in/nodes/Engineering%20Physics%20Notes.pdf
2. https://mrcet.com/downloads/digital_notes/HS/R20/Engineering%20Physics.pdf
3. NPTEL Course on Introduction To Electromagnetic Theory by Prof. Manoj Harbola, IIT Kanpur

4. NPTEL Course on Engineering Mechanics by Prof. Manoj Harbola, IIT Kanpur

Subject Name: Mathematics-II		**Subject Code: MAT022C202**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to enable students to achieve conceptual understanding and to retain the best traditions of traditional calculus.

Prerequisites: level Concepts of Mathematics I

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**List** the methodologies used for solving various equations	**BT 1**
CO 2	**Understand** essential tool of matrices and linear algebra in a comprehensive manner	**BT 2**
CO 3	**Utilize** the essential tools in the field of applied sciences and related fields.	**BT 3**
CO 4	**Analyze** and **evaluate** the qualitative behavior of solutions of systems of differential equations and interpret in the context of an underlying model.	**BT 4 & 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I.	**Matrices**	Linear Systems of Equations; Linear Independence; Rank of a Matrix; Determinant, Inverse of a matrix, rank-nullity theorem; System of linear equations; Symmetric, skew-symmetric and orthogonal matrices; Determinants; Eigenvalues and eigenvectors; Orthogonal transformation; Diagonalization of matrices; Cayley-Hamilton Theorem.	**16**
II.	**First order ordinary differential equations & Ordinary differential equations of higher orders**	Exact, linear and Bernoulli's equations. Equations not of first degree: equations solvable for p, equations solvable for y, equations solvable for x and Clairaut's type. Second order linear differential equations with variable coefficients: Euler-Cauchy equations, solution by variation of parameters; Power series solutions: Legendre's equations and Legendre polynomials, Frobenius method, Bessel's equation and Bessel's functions of the first kind and their properties.	**17**

III.	**Complex Variable – Differentiation**:	Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate; elementary analytic functions (exponential, trigonometric, logarithm) and their properties; Conformal mappings, Mobius transformations and their properties.	**16**
IV	**Complex Variable – Integration**:	Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using the Bromwich contour.	**17**
TOTAL			**66**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*22 NCH = 66 NCH	1*15 NCH = 15 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. A text book of Engineering Mathematics, Bali N. P. and Narayan Iyenger N., 9th Edition, 2016, Laxmi Publication.
2. Mathematical Methods for Physics and Engineering: A Comprehensive Guide, K. F. Riley, M. P. Hobson, 3rd Edition, 2006, Cambridge University Press
3. Reena Garg, Engineering Mathematics, Khanna Book Publishing Company, 2022.
4. Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021.

Reference Books:

1. Grewal B. S., Higher Engineering Mathematics, 43rd Edition, 2014, Khanna Publishers.
2. Das B. C. & Mukherjee B. N., Differential Calculus, 55th Edition, U. N. Dhur & Sons Pvt. Ltd.
3. Das B. C. & Mukherjee B. N, Integral Calculus, 57th Edition, U. N. Dhur & Sons Pvt. Ltd
4. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, John Wiley & Sons, 2006.
5. Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
6. W. E. Boyce and R. C. DiPrima, Elementary Differential Equations and Boundary Value Problems, 9th Edn., Wiley India, 2009.
7. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
8. S. L. Ross, Differential Equations, 3rd Ed., Wiley India, 1984.
9. E. A. Coddington, An Introduction to Ordinary Differential Equations, Prentice Hall

India, 1995.
10. E. L. Ince, Ordinary Differential Equations, Dover Publications, 1958.
11. J. W. Brown and R. V. Churchill, Complex Variables and Applications, 7th Ed., Mc-Graw Hill, 2004.
12. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Mathematics-I.
2. https://www.vidyalankar.org/gate/assets/docs/notes/maths.pdf

Subject Basic Electrical Engineering		**Subject Code: CSE022C215**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make students understand the basic electrical terminologies and familiarize them with the basic concepts of D.C., single-phase and three-phase A.C. networks.

Prerequisites: Basic concepts of D.C. networks of Class XII, Electromagnetic Induction and A.C. Fundamentals etc.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**ɹurse Outcome**	**Blooms Taxonomy Level**
CO 1	**Recall** the basic concepts of electrical engineering	**BT 1**
CO 2	**Understand** the concept behind basic electric and magnetic circuits.	**BT 2**
CO 3	**Apply** the working principles of electrical machines and power converters in real-life.	**BT 3**
CO 4	**Analyze** DC circuits using Ohm's Law and Kirchhoff's Laws, and understand the principles of electromagnetism	**CO 4**

Detailed Syllabus:

Modules	**Topic**	**Course Content**	**Hours**
I.	**DC Circuits**	Electrical Circuit Elements – The resistance element, the inductance element, the capacitance element. Voltage & Current source, practical & ideal voltage and current sources, source transformation. Kirchhoff's Laws, Analysis of simple circuits with DC excitation – series circuit, parallel circuit, voltage and current divider rule, star -delta conversion, Maxwells mesh current method, nodal voltage analysis ,	**12**

II.	**AC Circuits**	AC fundamentals – generation of alternating voltage, representation of sinusoidal waveform, concept of frequency, cycle, time period, instantaneous value, average value, peak value, RMS value, phasor representation. Single phase AC Circuits – analysis of single-phase AC circuits consisting of R-L-C parameters, apparent power, real power, reactive power, power factor and its significance. Analysis of R-C series circuit, R-L-C series circuit, analysis of AC parallel circuits	**20**
III.	**Electrical Machines:**	Principle of operation and construction of single-phase transformers. EMF equation, losses, efficiency and voltage regulation. DC Machines – Constructional details of a DC Machine; EMF Equation of a DC machine, Types of DC Machines, Applications of DC Generators, operation of a DC machine as a motor, Torque equation, importance of back emf, speed equation, speed regulation, starting a DC motor, types of DC Motor, applications of DC motors	**20**
IV.	**Electrical Installations:**	Electrical Power Supply System. Three phase four wire distribution system. Protection of electrical installations against overload, short circuit and earth fault. Protective devices for overload, short circuit, earth fault and electric shock – SFU, MCB, ELCB. Earthing – difference between neutral wire & earth wire, methods of earthing of domestic fittings and appliances. Types of wires, cables and wiring used in electrical installations.	**14**
TOTAL			**66**

Basic Electrical Engineering Lab Syllabus

Basic Electrical Engineering	**Subject Code: CSE022C215**	
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

Lab	Experiments	Hours
I	To verify Thevenin's Theorem for DC network	**2**
II	To verify Maximum Power Transfer Theorem for DC network	**2**
III	Study of R-L-C Series circuit and determine R,L,C, cos Φ,P and Q and draw the phasor diagram	**2**
IV	Study of R-L-C Parallel circuit and determine R,L,C,cos Φ,P and Q and draw the phasor diagram	**2**
V	Calibration of a milli-ammeter as a voltmeter.	**2**
VI	To determine the ohmic and effective resistance (armature winding)	**2**
VII	To study the characteristics of a filament lamp	**2**
VIII	To measure the power in a single-phase load using one wattmeter	**2**
IX	To measure the insulation resistance using Megger	**2**
X	Demonstration of house wiring	**2**
	TOTAL	**20**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. A Text Book of Electrical Technology, Thereja, B.L., 1st Edition revised, 2008, S Chand & Company Ltd. Ram Nagar; New Delhi.
2. Basic Electrical Engineering, D. P. Kothari, I. J.Nagrath, 3rd Edition, 2009, Tata McGraw-Hill

Reference Books:

1. D. C. Kulshreshtha, *Basic Electrical Engineering*, 1st Edition, 2009, McGraw-Hill
2. E. Hughes, *Electrical and Electronics Technology,* 10th Edition, 2011, Pearson Publication

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Basic%20Electrical%20Engineering%20R-20.pdf
2. https://www.cet.edu.in/noticefiles/231_BASIC_ELECTRICAL_ENGG-min.pdf
3. NPTEL Course on Basic Electrical Circuits by Prof. Nagendra Krishnapura, IITM
4. NPTEL Course on Fundamentals of Electrical Engineering by Prof. Debapriya Das, IIT, Kharagpur

Subject Name: Engineering Mechanics		**Subject Code:** CEE022C217
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Equilibrium of System of Forces**	Force Systems Basic concepts, equilibrium in 2-D & 3-D; Rigid Body equilibrium; System of Forces, Coplanar Concurrent Forces, Components in Space – Resultant-Moment of Forces and its Application; Couples and Resultant of Force System. Equilibrium of System of Forces, Free body diagrams, Equations of Equilibrium of Coplanar Systems and Spatial Systems; Static Indeterminacy	**15**
II.	**Analysis Of Structure**	Two force members, Analysis of plane trusses by Method of joints Analysis of plane trusses by method of section, Analysis of plane frames, Cables subjected to point load multi force member.	**15**
III.	**Centroid, Centre Of Gravity And Moment Of Inertia**	Centroid of simple figures from first principle, Centroid of composite sections; Centre of Gravity and its implications. Area moment of inertia- Definition, Moment of inertia of plane sections from first principles, Theorems of moment of inertia, Moment of inertia of standard sections and composite sections; Mass moment inertia of circular plate, Cylinder, Cone, Sphere, Hook.	**15**
IV	**Friction, Virtual Work And Energy Method**	Types of friction, Limiting friction, Laws of Friction, Static and Dynamic Friction; Motion of Bodies, wedge friction, screw jack. Virtual displacements, principle of virtual work for particle and ideal system of rigid bodies, degrees of freedom. Active force diagram, systems with friction, mechanical efficiency. Conservative forces and potential energy (elastic and gravitational), energy equation for equilibrium. Applications of energy method for equilibrium. Stability of equilibrium.	**15**
TOTAL			**60**

Textbooks:

1. F. P. Beer and E. R. Johnston (2011), Vector Mechanics for Engineers, Vol I - Statics, Vol II, – Dynamics, 9th Ed, Tata McGraw Hill
2. R. C. Hibbler (2006), Engineering Mechanics: Principles of Statics

and Dynamics, Pearson Press.
3. Timoshenko S, Young H.D. and Rao J.V, Engineering Mechanics, Tata McGraw Hill Publications.

Subject Name: English for Technical Writing		**Subject Code: CEN982A203**
L-T-P-C – 2-0-0-2	**Credit Units: 02**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to provide learning environment to practice listening, speaking, reading and writing skills, to assist the students to carry on the tasks and activities through guided instructions and materials and to effectively integrate English language learning with employability skills and training.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the various forms of communication	**BT 1**
CO 2	**Understand** basic proficiency in English.	**BT 2**
CO 3	**Develop** reading and listening comprehension, writing and speaking skills.	**BT 3**
CO 4	**Analyze** the type of communication	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Hours**
I	**Vocabulary Building**	The concept of Word Formation, Root words from foreign languages and their use in English, Acquaintance with prefixes and suffixes from foreign languages in English to form derivatives, Synonyms, antonyms, and standard abbreviations.	**10**
II	**Basic Writing Skills**	Sentence Structures, Use of phrases and clauses in sentences, Importance of proper punctuation, creating coherence, organizing principles of paragraphs in documents, Techniques for writing precisely, Subject-verb agreement, Noun-pronoun agreement, Misplaced modifiers, Articles, Prepositions, Redundancies, Clichés	**10**
III	**Writing Practices**	Nature and Style of sensible Writing, Describing, Defining 1.3. Classifying, providing examples or	**10**

		evidence, Writing introduction and conclusion, Comprehension, Précis Writing, Essay Writing.	
IV	**Oral Communication**	Listening Comprehension, Pronunciation, Intonation, Stress and Rhythm, Common Everyday Situations: Conversations and Dialogues, Communication at Workplace, Interviews, Formal Presentations	**10**
TOTAL			**40**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2*20 NCH = 40 NCH	-	20 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. Effective Communication Skills. Kul Bhushan Kumar, 2022, Khanna Book Publishing
2. Practical English Usage, Michael Swan. 1995, OUP

Reference Books:

1. F.T. Wood, *Remedial English Grammar.,* 2007, Macmillan.
2. William Zinsser, *On Writing Well*, 2001, Harper Resource Book.
3. Liz Hamp-Lyons and Ben Heasly, *Study Writing*, 2006,
4. Sanjay Kumar and PushpLata, *Communication Skills*, 2011, Oxford University Press.

Additional Readings:

1. AICTE's Prescribed Textbook: English (with Lab Manual), Khanna Book Publishing Co., https://khannabooks.com/index.php?route=product/product&path=99_105&product_id=480
2. NPTEL Course on English Language for Competitive Exams by Prof. by Aysha Iqbal, IIT, Madras
3. NPTEL Course on Technical English for Engineers by Prof. by Aysha Iqbal, IIT, Madras

Subject Name: Design Thinking & Ideation Lab		**Subject Code:** COD022S216
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

To develop students' creative thinking, innovation, and problem-solving abilities through the Design Thinking process, enabling them to identify real-world engineering challenges, generate sustainable solutions, develop prototypes, and communicate innovative ideas effectively.**Prerequisites:** None

Course Outcomes:

Upon successful completion of the course, students will be able to:

CO	Course Outcome	Bloom's Level
CO1	**Explain the principles of design thinking, creativity, ideation and innovation in solving engineering problems.**	**BT1–BT2**
CO2	**Apply empathy, stakeholder analysis and creative thinking techniques to identify problems and generate innovative engineering concepts.**	**BT3**
CO3	**Develop, test and refine engineering prototypes by integrating sustainability, technical feasibility and user feedback.**	**BT4–BT5**
CO4	**Design, document and present innovative engineering solutions considering entrepreneurship, intellectual property and scalability.**	**BT5–BT6**

Detailed Syllabus:

Module	Topics	Course Contents	Hours
I	**Foundations of Design Thinking, Creativity and Problem Identification**	Introduction to creativity, innovation and ideation; importance of Design Thinking in engineering; stages of Design Thinking (Empathize, Define, Ideate, Prototype, Test); learning styles and creative thinking; empathy, stakeholder analysis, field observation, need identification; understanding societal, environmental and sustainability challenges; formulation of engineering problem statements.	**5**
II	**Idea Generation and Concept Development**	Brainstorming techniques, mind mapping, SCAMPER and other creative tools; problem-solving approaches; idea screening and feasibility analysis; technical, economic and environmental considerations; customer-centric design; user experience; engineering concept development; sustainable and resilient design principles; selection of suitable materials and methods.	**5**
III	**Prototype Development, Testing and Validation**	Engineering product design process; prototype development (low-fidelity and working models); rapid prototyping techniques; testing and validation; user feedback; ergonomic considerations; iterative	**6**

Module	Topics	Course Contents	Hours
		design improvement; group activities, teamwork and appreciation of individual differences; documentation of prototype development.	
IV	**Innovation, Entrepreneurship and Final Project**	Technical documentation and report writing; presentation and pitching skills; introduction to entrepreneurship and startup ecosystem; intellectual property rights (IPR); techno-economic feasibility; scalability of engineering solutions; final prototype demonstration; peer review; solving real-life civil engineering problems using Design Thinking and innovative solutions.	**6**

Suggested Text Books

- E. Balaguruswamy, *Developing Thinking Skills (The Way to Success)*, Khanna Publishing House, 2022.
- Iñigo Cuiñas & Manuel José Fernández Iglesias, *Design Thinking for Engineering: A Practical Guide*, Institution of Engineering and Technology, 2023.
- Christian Müller-Roterberg, *Design Thinking For Dummies*, Wiley, 2020.
- Idris Mootee, *Design Thinking for Strategic Innovation*, Adams Media, 2014.
- Chris Hackett, *The Big Book of Maker Skills*, Weldon Owen, 2018.

SEMESTER -III

DETAILED SYLLABUS OF 3rd SEMESTER

Subject Name: Solid Mechanics		**Subject Code: CE022C301**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objective of this Course is to introduce to continuum mechanics and material modelling of engineering materials based on first energy principles: deformation and strain; momentum balance, stress and stress states; elasticity and elasticity bounds; plasticity and yield design

Prerequisites: Physics and Mathematics

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the concepts of stress and strain	**BT 1**
CO 2	Understand the concepts of bending moment and shear stress and develop BM and SF diagrams	**BT 2**
CO 3	Apply concept of bending moment to calculate bending stress, section modulus	**BT 3**
CO 4	Apply concept of shear stress and and torsion to calculate stresses	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I.	**Stresses and Strains**	Concept of stress and strain, St. Venant's principle, stress and strain diagram, Elasticity and plasticity – Types of stresses and strains, Hooke's law – stress – strain diagram for mild steel – Working stress – Factor of safety – Lateral strain, Poisson's ratio and volumetric strain – Elastic moduli and the relationship between them – Bars of varying section – composite bars – Temperature stresses. Strain Energy – Resilience – Gradual, sudden, impact and shock loadings – simple applications. Compound Stresses and Strains- Two-dimensional system, stress at a point on a plane, principal stresses and principal planes, Mohr circle of stress, ellipse of	**12**
II.	**Bending moment and Shear Force Diagrams**	Bending moment (BM) and shear force (SF) diagrams.BM and SF diagrams for cantilevers simply supported and fixed beams with or without overhangs. Calculation of maximum BM and SF and the point of contra flexure under concentrated loads, uniformly distributed loads over the whole span or part of span, combination of concentrated loads (two or three) and uniformly distributed loads, uniformly varying loads, application of moments.	**11**
III.	**Flexural Stresses-Theory of simple bending**	– Assumptions – Derivation of bending equation: M/I = f/y = E/R - Neutral axis – Determination of bending stresses – Section modulus of rectangular and circular sections (Solid and Hollow), I,T, Angle and Channel sections – Design of simple beam sections.	**12**

IV.	**Shear Stresses and Torsion**	Shear Stresses- Derivation of formula – Shear stress distribution across various beam sections like rectangular, circular, triangular, I, T angle sections. Torsion- Derivation of torsion equation and its assumptions. Applications of the equation of the hollow and solid circular shafts, torsional rigidity, Combined torsion and bending of circular shafts, principal stress and maximum shear stresses under combined loading of bending and torsion. Analysis of close-coiled-helical springs.	**10**
TOTAL			**45**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH	NA	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

1. D.S. Bedi, "Strength of Materials", Khanna Book Publishing Co.
2. AICTE Prescribed Textbook: Physics (Introduction to Mechanics), Bhattarchaya, A.B., Khanna Book Publishing Co., 2023.
3. Timoshenko, S. and Young, D. H., "Elements of Strength of Materials", DVNC, New York, USA.
4. Kazmi, S. M. A., "Solid Mechanics" TMH, Delhi, India.
5. Hibbeler, R. C. Mechanics of Materials. 6th ed. East Rutherford, NJ: Pearson Prentice Hall, 2004
6. Crandall, S. H., N. C. Dahl, and T. J. Lardner. An Introduction to the Mechanics of Solids. 2nd ed. New York, NY: McGraw Hill, 1979
7. Laboratory Manual of Testing Materials - William Kendrick Hall
8. Mechanics of Materials - Ferdinand P. Beer, E. Russel Jhonston Jr., John T. DEwolf – TMH 2002.
9. Strength of Materials by R. Subramanian, Oxford University Press, New Delhi

Reference Books:

3. D.S. Prakash Rao, *Structural analysis: Unified approach*, Universities Press.
4. C.H. Norris, J.B. Wilbur, S. Utku, *Elementary Structural Analysis*, Tata McGraw Hill.
5. L. S. Negi and R. S. Jangid, *Structural Analysis*, Tata McGraw Hill, New Delhi.
6. C.K. Wang, *Intermediate Structural Analysis*, McGraw Hill, International Edition.
7. William M.C. McKenzie, *Examples in Structural Analysis*, CRC Press.

Subject Name: Strength of Mechanics Lab Subject Code: CEE022C311

L-T-P-C – 0-0-2-1 Credit Units: 01 Scheme of Evaluation: Lab

List of Experiments:

1. Tension test
2. Bending tests on simply supported beam and Cantilever beam.
3. Compression test on concrete
4. Impact test
5. Shear test
6. Investigation of Hook's law that is the proportional relation between force and stretching in elastic deformation,
7. Determination of torsion and deflection,
8. Measurement of forces on supports in statically determinate beam,
9. Determination of shear forces in beams,
10. Determination of bending moments in beams,
11. Measurement of deflections in statically determinate beam,
12. Measurement of strain in a bar
13. Bend test steel bar;
14. Yield/tensile strength of steel bar;

Subject Name: Building Planning & CAD		Subject Code: CEE022C303
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objective of this course is to equip students with the knowledge and skills necessary to effectively plan, design, and implement various aspects of building construction and services. Students will learn to apply principles of building planning, create detailed construction drawings, and manage essential building services.

Prerequisites: Engineering Graphics & Design and Basic Computer

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** drafting commands of AutoCAD.	**BT 1**
CO 2	**Explain** the concept of the built environment and apply principles from the National Building Code and green building standards to the planning and design of residential and public buildings.	**BT 2**
CO 3	**Develop** the ability to prepare detailed construction drawings, including plans, elevations, sections, and site plans, for various types of buildings such as residential, educational, healthcare, and entertainment structures.	**BT 3**
CO 4	**Examine** efficient water supply, drainage, and solid waste management systems for different building types, incorporating sustainable practices like wet and dry waste segregation and vermi-composting.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I.	**Principles of Residential and Public Buildings**	Concept of built environment and its application in planning. Recommendation of National building code., Green building, Introduction-Benefits, National priorities, rating system, check list, Site selection and planning, Water efficiency, Energy efficiency, Materials, Indoor environmental quality, Innovation and design process.	**8**
II.	**Planning of Building**	Preparation of constructional details and drawings-plan, elevation, section, site plan, foundation plan, terrace plan, waterproofing treatment, typical door and window. Planning of building such as • Residential building –Load bearing structure, RCC framed structure. Building for Education – school, college. Library • Building for health –Dispensary, Hospital Industrial structure • Building for entertainment-Theatre, club house, sports club. Other structure - Office, Hostel, Guest house.	**7**
III.	**Building's Water Supply and Drainage & Solid Waste Collection and Disposal System.**	Design of water supply, waste water and storm water collection system for various types of buildings. Pumps and Pump House. Wet and dry solid waste segregation, Vermi-composting etc. Provision of Chutes. Accessibility in public Sanitation Systems.	**7**

IV.	**Fire Protection System** **Elevators** **Heating Ventilation and Air Conditioning** **Building Management System**	**Fire Protection System**: -Introduction, Fire protection, requirement of water quantity estimation. Systems of firefighting external and internal. Wet and dry risers, smoke alarm, Sprinkler system. Safety corridors in High- rise structures. **Elevators**: -Introduction, types of elevators. Essential features of lifts its size and requirement of minimum numbers, norms for safety doors, Operation and maintenance, Safety norms. Control systems, electrical requirement, and generator back-up, Escalators in Industry and in malls-multiplex. Design of Accessible Circulation System for differently abled publics. **Heating Ventilation and Air Conditioning**: -Ventilation, functional requirement, Heat balance system of ventilation, General rules and regulations in artificial ventilation system, Central air conditioning: - ducting and glass claddings. Operation and maintenance **Building Management System**: -Security Guard's Cabin, Postage collection boxes, Parking space.	**8**
TOTAL			**30**

Subject Name: Fluid Mechanics		**Subject Code: CEE022C304**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: TP**

The objective of this course is to introduce the concepts of fluid mechanics useful in Civil Engineering applications. The course provides a first level exposure to the students to fluid statics, kinematics and dynamics. Measurement of pressure, computations of hydrostatic forces on structural components and the concepts of Buoyancy all find useful applications in many engineering problems. A training to analyse engineering problems involving fluids – such as those dealing with pipe flow, open channel flow, jets, turbines and pumps, dams and spillways, culverts, river and groundwater flow - with a mechanistic perspective is essential for the civil engineering students. The topics included in this course are aimed to prepare a student to build a good fundamental background useful in the application-intensive courses covering hydraulics, hydraulic machinery and hydrology in later semesters.

Module 1: Basic Concepts and Definitions – Distinction between a fluid and a solid; Density, Specific weight, Specific gravity, Kinematic and dynamic viscosity; variation of viscosity withtemperature, Newton law of viscosity; vapour pressure, boiling point, cavitation; surface tension, capillarity, Bulk modulus of elasticity, compressibility.

Module 2: Fluid Statics - Fluid Pressure: Pressure at a point, Pascals law, pressure variation with temperature, density and altitude. Piezometer, U-Tube Manometer, Single Column Manometer, U-Tube Differential Manometer, Micromanometers. pressure gauges, Hydrostatic pressure and force: horizontal, vertical and inclined surfaces. Buoyancy and stability of floatingbodies.

Module 3: Fluid Kinematics- Classification of fluid flow : steady and unsteady flow; uniform and non-uniform flow; laminar and turbulent flow; rotational and irrotational flow; compressible and incompressible flow; ideal and real fluid flow; one, two and three dimensional flows; Stream line, path line, streak line and stream tube; stream function, velocitypotential function. One-, two- and three -dimensional continuity equations in Cartesian coordinates

Module 4: Fluid Dynamics- Surface and body forces; Equations of motion - Euler's equation;Bernoulli's equation – derivation; Energy Principle; Practical applications of Bernoulli's equation : venturimeter, orifice meter and pitot tube; Momentum principle; Forces exerted by fluid flow on pipe bend; Vortex Flow – Free and Forced; Dimensional Analysis and Dynamic Similitude - Definitions of Reynolds Number, Froude Number, Mach Number, Weber *Number and Euler Number; Buckingham's* π-Theorem.

Subject Name: Fluid Mechanics Lab		**Subject Code: CEE022C314**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: TP**

Lab Experiments

1. Measurement of viscosity
2. Study of Pressure Measuring Devices
3. Stability of Floating Body
4. Hydrostatics Force on Flat Surfaces/Curved Surfaces
5. Verification of Bernoulli's Theorem
6. Venturimeter
7. Orifice meter

8. Impacts of jets
9. Flow Visualisation -Ideal Flow
10. Length of establishment of flow
11. Velocity distribution in pipes
12. Laminar Flow

Text/Reference Books:

1. Fluid Mechanics, Sadhu Singh, Khanna Book Publishing Co., New Delhi
2. Fluid Machinery (Hydraulic Machines), Sadhu Sigh, Khanna Book Publishing Co.,
3. Fluid Mechanics and Machinery, C.S.P.Ojha, R. Berndtsson and P. N. Chadramouli,Oxford University Press, 2010
4. Hydraulics and Fluid Mechanics, P M Modi and S M Seth, Standard Book House
5. Theory and Applications of Fluid Mechanics, K. Subramanya, Tata McGraw Hill
6. Fluid Mechanics with Engineering Applications, R.L. Daugherty, J.B. Franzini and E.J.Finnemore, International Student Edition, Mc Graw Hill.

Course Outcomes:

At the end of the course, the student will be able to:

- Understand the broad principles of fluid statics, kinematics and dynamics
- Understand definitions of the basic terms used in fluid mechanics
- Understand classifications of fluid flow
- Be able to apply the continuity, momentum and energy principles
- Be able to apply dimensional analysis.

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Subject Name: Concrete Technology		**Subject Code: CEE022C305**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

To understand the fundamentals of concrete and its ingredients and to be able to design and concrete mixes of different grades

Prerequisites: Concepts of +2 level Physics

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define the functional role of all ingredients of concrete and their use for normal and special purpose concrete.**	**BT 1**
CO 2	**Apply the principle of sustainability for the utilization of waste, novel and innovative materials for use in concrete.**	**BT 2**
CO 3	**Formulate concrete mix for normal and special purpose concrete.**	**BT 3**
CO 4	**Use of various non-destructive testing procedure for evaluation of concrete properties.**	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course Content**	**Periods**
I.	*Properties of ingredients*	Properties of coarse and fine aggregates and their influence on concrete, types of cement and their use, Grades of ordinary Portland cement, Portland pozzolana cement, rapid hardening Portland cement, hydrophobic cement, low heat Portland cement and sulphate resisting Portland cement as per relevant I.S. codes. Types of aggregates and their properties. Testing of aggregates as per relevant IS Codes.	**8**
	Properties of different types of concrete	Concrete for structural work, light weight concrete, high density concrete, biological concrete, workability, durability and strength requirements, effect of w/c ratio on properties of fresh and hardened concrete, acceptability criteria, laboratory testing of fresh and hardened concrete, Fire resistant properties of hardened concrete.	
II.	**Concreting methods**	Process of manufacturing of concrete, transportation, placing, compaction and curing of concrete. Extreme weather concreting, special concreting methods, vacuum dewatering– underwater concrete, special form work., Plum	**7**

		Concrete, Self-Compacting Concrete	
	Admixtures	Plasticizers, Retarders, Accelerators and other Admixtures, Test on Admixtures, Chemistry and Compatibility with concrete. GGBS fly Ash, Metakaolin, Silica Fumes, crush sand,	
III.	**Ready mix concrete**	Requirements of ready-mix concrete, properties of RMC, transit mixer details, Automation, instrumentation and Layout of RMC plant.	**8**
	Concrete mix design	Mix Design for compressive strength by I.S. methods, road note method, British method, ACI Method, Mix design for flexural strength.	
IV.	**Concrete for repairs and rehabilitation of structures**	High Performance concrete, Polymer Concrete, Fiber Reinforced Concrete, Light weight concrete and its manufacture, Polymer Impregnated Cement Concrete, Polymer Modified cement concrete and Ferro Cement, Special Tests for concrete used for repairs and rehabilitation	**7**
	Non-destructive testing of concrete	Rebound hammer test, Ultrasonic pulse velocity test, Magnetic particle testing, Liquid penetration testing, Visual testing, Laser Testing methods, Leak Testing, Impact echo test, carbonation test, Half cell potentiometer and corrosion of steel, Core test and relevant provisions of I.S. codes.	
TOTAL			**30**

Concrete Technology Lab Syllabus

Total Lab Hours for the semester = 30 (2 hours per week) **Subject Code: CEE022C315**

L-T-P-C – 0-0-2-1 **Credit Units: 01** **Scheme of Evaluation: P**

Minimum 10 Laboratory experiments based on the following-

Objective: To determine the various parameters of road building materials and understand their physical properties.

1. Sieve analysis
2. Setting Time Test
3. Soundness Test
4. Workability
5. Compressive Strength Test
6. Concrete Mixing for Grade M25

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2 * 15 NCH = 30 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

1. **M.L. Gambhir, Concrete Technology, McGraw Hill Book Company, Fifth Edition, 2017. (ISBN- 1259062554, 978-1259062551).**
2. **M.S. Shetty, Concrete Technology, Theory and Practice, S. Chand Publication, Sixth Edition, 2018. (ISBN- 9788121900034,978-8121900034)**
3. **B.L. Gupta and A. Gupta, Concrete Technology, Jain Book Agency, 2013. (ISBN-8180140407, 978-8180140402).**

Recommended Reading

1. **A.R. Santhakumar, Concrete Technology, Oxford University Press, New Delhi, 2018. (ISBN- 9780195671537, 978-0195671537).**

2. **A.M. Neville, Properties of Concrete, Pearson Publication, London, 2012. (ISBN-978- 0273755807, 9780273755807).**

3. **IS 10262-(2009) Recommended Guidelines for Concrete Mix Design, Bureau of Indian Standards, New Delhi, 2009.**
4. **IS10262 (2009), Mix Design**

5. **IS269 (2015), Ordinary Portland Cement (33 Grade).**

6. **IS12269 (2013), Ordinary Portland Cement (53 Grade).**

7. **IS650 (1991), Specification of Standard Sand. 8. IS383 (1970), Specification for Coarse and Fine aggregate.**

Subject Name: Mathematics III	**Subject Code: MAT022C306**
L-T-P-C –3-0-0-3 **Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The course aims to familiarize students with fundamental concepts of mathematics that they will encounter during their studies in the industry and engineering fields, as well as to help them efficiently understand new developments and breakthroughs in engineering and technology.

Prerequisites: (i) Physics (ii) Mathematics I and II

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the basic mathematical definitions and properties required in engineering	BT 1
CO 2	Understand the concepts of vectors, differential equations, important series and transformations, matrices	BT 2
CO 3	Apply these concepts in solving problems	BT 3
CO 4	Analyze these concepts to independently do problems and relate it to real life applications	BT 4

Detailed Syllabus:

Modules	**Topics**	**Course Content**	**Periods**

I.	**Basic concepts of vector calculus**	Scalar and vector point function, differential operator, gradient, directional derivative, physical meaning of gradient, divergence, curl and Laplacian with their properties. Line Integrals, Surface Integral, Volume integral, Green's theorem, Gauss' theorem and Stoke's theorem & its applications.	**18**
II.	**Fourier series and partial differential equation**	Definition of Fourier series, Orthogonal and orthonormal functions Fourier series with arbitrary period, in particular periodic function with period 2π Fourier series of even and odd function Half range Fourier series. Second order PDE of mathematical physics (Heat, wave and Laplace equation, one dimensional with standard boundary conditions) Solution by separation of variable method using Fourier series.	**18**
III.	**Laplace transforms and applications**	Introduction, Definition of the Laplace transform, Useful properties of Laplace transform (without proof): Linearity, Frist shifting theorem, Multiplication and division by t, transforms of derivatives and integrals, Heaviside unit step function, Dirac's delta function, second shifting theorem, Laplace transform of Periodic function. Inverse Laplace transform using partial fraction and Convolution theorem (without proof). Application to solve initial and boundary value problem involving ordinary differential equations with one dependent and constant coefficient.	**18**
IV.	**Matrices**	Eigen values Eigen vectors of square matrix, Cayley Hamilton's theorem and function of square matrix, Diagonalization of square matrix. Minimal Polynomial and Minimal	**18**

		Equation of a Matrix. Derogatory and Non-Derogatory Matrices.	
TOTAL			**72**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 18 NCH = 72 NCH	**NIL**	**48 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)**

Text Books:

1. AICTE's Prescribed Textbook: Mathematics-I (Calculus & Linear Algebra), Reena Garg, Khanna Book Publishing Co., New Delhi, 2023.
2. Reena Garg, Engineering Mathematics, Khanna Book Publishing Company, 2022.
3. Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021.
4. Kreyszing E., "Advanced Engineering Mathematics", John Wiley & Sons, Singapore, Int.Student Ed. 1995. (ISBN 8126554231)

Reference Books:

1. Wiley C. R., "Advanced Engineering Mathematics", McGraw Hill Inc., New York Ed.1993.
2. O'Neel Peter., "Advanced Engg. Mathematics", Thompson, Singapore, Ind. Ed. 2002.
3. Greenbar Michael D., "Advanced Engg. Mathematics", Pearson, Singapore, Ind. Ed.
4. Ramana D. V., "Higher Engg. Mathematics", The MaGraw-Hill Inc., New Delhi, 2007. (ISBN 007063419X)
5. Marsden J. E., Tromba A., Weinstein A., "Basic multivariable calculus", Springer, 1993. (ISBN 354097976X)
6. A. R. Vasishtha, A. K. Vasishtha, "Matrices", Krishna Prakashan Media, 1991. (ISBN 8182837294)

Subject Name: IKS	**Subject Code:**IKS022C305
L-T-P-C –2-0-0-2 Credit Units: 02	**Scheme of Evaluation: T**

This Foundation course presents a consolidated overview of the streams of IKS relevant to the UG program within a 2-credit framework. It enables students to explore the most fundamental ideas that have shaped Indian Knowledge Traditions, including Ayurveda and Yoga as living systems of health and wellbeing, over the centuries.

Course Outcomes (CO)

On completion of this course, students will be able to:

CO	Course Outcome Contents	BT Level	SDG Mapping
CO1	Recall the rich heritage of Indian knowledge systems, including Ayurveda and Yoga	BT Level 1	SDG 3, SDG 4
CO2	Describe the contribution of Indian knowledge systems — including traditional health sciences — to the world	BT Level 2	SDG 3, SDG 4
CO3	Demonstrate knowledge of the sociocultural, ethnolinguistic, and ecological diversity that constitutes the soul of Bharatvarsha	BT Level 2	SDG 10, SDG 15

CO4	Apply traditional knowledge, techniques, and wellbeing practices (Yoga, Ayurvedic lifestyle) in day-to-day life	BT Level 3	SDG 3
CO5	Distinguish and critically evaluate knowledge traditions that originated in the Indian subcontinent	BT Level 3	SDG 4, SDG 11

SDG Legend:

SDG 3 – Good Health & Well-being | SDG 4 – Quality Education | SDG 10 – Reduced Inequalities | SDG 11 – Sustainable Cities & Communities | SDG 15 – Life on Land

Module-wise Course Contents

Module	Title	Course Contents	Periods
I	**Foundations of IKS & Ancient Bharat Varsha**	• Introduction to IKS: definition, scope, and importance of Indigenous/Traditional Knowledge • People of Ancient Bharat Varsha; the great Himalayas and rivers • Civilizations of the Sindhu-Ganga Valley and the Brahmaputra Valley; coastal plains • Forests and Minerals; ancient traditional wisdom about nature and climate • Agriculture • Textile	**8**
II	**Indian Heritage of Knowledge, Language & Arts**	• The Vedas and Vedangas; the Sastras; the Ramayana and Mahabharata • Madhav Kandali's Ramayan and Srimanta Sankardev's Dasam Skandha Bhagavat • Linguistics; Varnamala; Nirukta, Vyakarana, Chandas; Paninian Grammar (brief) • Natyasastra and Gandharva-Veda; folk music and dance forms of the Northeast	**8**
III	**Indian Science, Technology & Astronomy**	• Number System, Algebra, Geometry; origin of the Decimal system; Zero and Infinity; Sulba-sutras • Contribution of Brahmagupta and Sridhar Acharya • Indian Astronomy: planetary motion, velocity of light, eclipses, Navagrahas; Aryabhata and Nilakantha • Indian metalworks: mining, metal types, smelting techniques, with reference to NE India	**7**
IV	**Ayurveda, Yoga & Contribution to Health Sciences**	• Traditional Indigenous systems of medicine: Ayurveda and Yoga • Elements of Ayurveda: Gunas and Doshas, Pancha Mahabhuta, Sapta-dhatu; concept of disease • Ayurvedic lifestyle practices: Dinacharya and Ritucharya; hospitals in Ancient India • Yoga as an integral IKS tradition: principles and relevance to holistic wellbeing	**7**

Module	Title	Course Contents	Periods
		• Ayurveda and Yoga as India's gift to the modern world	
	Total		**30**

Textbooks

- Mahadevan, B., Bhat, Vinayak Rajat, Nagendra Pavan RN. (2022). Introduction to Indian Knowledge System: Concepts and Applications. PHI Learning Private Ltd.
- Mukul Chandra Bora. Foundations of Bharatiya Knowledge System. Khanna Book Publishing.

Reference Books

- Baladev Upadhyaya, Samskrta Śāstrom ka Itihās, Chowkhambha, Varanasi, 2010.
- D. M. Bose, S. N. Sen and B. V. Subbarayappa, Eds., A Concise History of Science in India, 2nd Ed., Universities Press, Hyderabad, 2010.
- Astāngahrdaya, Vol. I, Sūtrasthāna and Śarīrasthāna, Translated by K. R. Srikantha Murthy, Krishnadas Academy, Varanasi, 1991.
- Dharampal, The Beautiful Tree: Indian Indigenous Education in the Eighteenth Century, Rashtrotthana Sahitya, Bengaluru, 2021.
- J. K. Bajaj and M. D. Srinivas, Indian Economy and Polity in Eighteenth-century Chengalpattu, in J. K. Bajaj ed., Indian Economy and Polity, Centre for Policy Studies, Chennai, 1995, pp. 63–84.

Experiential Learning Sessions

- Field Visits: historical sites, museums, traditional craft centers relevant to IKS
- Interactive Sessions: discussions with experts/practitioners across IKS domains
- Online Lecture Series: sessions by distinguished experts in the field of IKS
- Hands-on Activities: traditional arts, crafts, music, dance, and agriculture
- Practical Demonstrations: workshops on Yoga, Ayurveda, and Vastu Shastra

SEMESTER -IV

Subject Name: Structural Analysis I		**Subject Code: CEE022401**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objective of the course is to provide students with a comprehensive understanding of the fundamental principles and methods of structural analysis. The course aims to develop their ability to apply analytical techniques for determining internal forces, and deflections in structures under various loading conditions. Through the study of beams, columns, trusses, arches, suspension bridges students will be prepared to analyze and design real-world structural systems.

Prerequisites: Solid mechanics

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define structure types, loads, and basic theorems in structural analysis	**BT 1**
CO 2	Understand buckling concepts and theories for struts and columns	**BT 2**
CO 3	Apply influence lines to analyze beams, trusses, arches, and suspension bridges	**BT 3**
CO 4	Analyze indeterminate structures using stiffness and flexibility methods.	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course Content**	**Periods**
I.	**Fundamentals of Structural Analysis (5 Hours)**	Classification of Structures-based on geometry, based on stiffness, based on materials of construction, Basic Requirements of Structures-Structural stability, Support Systems, Intro duction to Statically Determinate and Indeterminate structures. Free body diagram.	**05**
II.	**Deflection of Beams and Frames**	Relation between bending moment and deflection., **Double integration method**, **Macauley's method**, **Moment-area theorems**, **Conjugate beam method**, **Strain energy method** (Castigliano's Theorem – I and	**20**

		II), Deflection of determinate beams and pin-jointed frames.	
III.	**Analysis of Perfect Frames**	Types of trusses: perfect, deficient, and redundant, Analysis of trusses using: Method of joints: Method of sections: Graphical method (introduction): Application to plane trusses (Warren, Pratt, Howe).	**15**
IV.	**Arches and Cables**	Three-hinged and two-hinged arches: analysis for bending moment, shear, and thrust, Eddy's theorem, Suspension cables and parabolic cables: tension and shape of cables, ILD for determinate structures for reactions at supports.	**15**
TOTAL			**55**

Text book(s):

1. Structural Analysis (Vol – I): S.S. Bhavikatti, Vikas Publication.

2. Theory of Structures: S. Ramamrutham and R. Narayan, Dhanpat Rai & Sons.

Reference Book(s):

1. Analysis and Design of Structures: Vazirani, Ratwani & Duggal: Khanna Publisher.

2. R.C. Hibbeler, Structural Analysis, Pearson Education.

3. C.S. Reddy, Basic Structural Analysis, Second Edition, Tata McGraw Hill.

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Subject Name: Hydraulic Engineering	Subject Code: CEE022C402	
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

To introduce the students to various hydraulic engineering problems like open channel flows and hydraulic machines. At the completion of the course, the student should be able to relate the theory and practice of problems in hydraulic engineering

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand and analyze fluid flow behavior in pipes and open channels, including laminar and turbulent flow characteristics, boundary layer concepts, and flow measurement techniques.**	**BT 1**
CO 2	**Apply principles of dimensional analysis and similitude for hydraulic modeling and solve problems related to fluid flow using appropriate theoretical and empirical methods.**	**BT 2**
CO 3	**Evaluate uniform and non-uniform flow conditions in open channels, including flow classification, hydraulic jump phenomena, and energy and momentum principles.**	**BT 3**
CO 4	**Design and analyze pipe networks, hydraulic structures, and flow control devices, and utilize computational tools to simulate fluid flow and optimize water resource systems.**	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours

I.	Fundamentals of Soil Mechanics and Soil Properties	Laminar flow through circular pipes, annulus, and parallel plates, Stoke's law, measurement of viscosity, Reynolds experiment, transition from laminar to turbulent flow, definition of turbulence, scale and intensity of turbulence, causes of turbulence, instability and mechanism of turbulence, effect of turbulent flow in pipes, Reynolds stresses, semi-empirical theories of turbulence, Prandtl's mixing length theory, universal velocity distribution equation, resistance to flow of fluid in smooth and rough pipes, Moody's diagram.	**16**
II.	Boundary Layer Theory and Dimensional Analysis	Assumption and concept of boundary layer theory, boundary-layer thickness, displacement thickness, momentum thickness, energy thickness, laminar and turbulent boundary layers on a flat plate, laminar sub-layer, smooth and rough boundaries, local and average friction coefficients, separation and control of boundary layers, dimensional homogeneity, Rayleigh method, Buckingham's Pi method and other methods, dimensionless groups, similitude, model studies, types of models, application of dimensional analysis and model studies to fluid flow problems.	**17**
III.	Open Channel Flow and Hydraulic Phenomena	Introduction to open channel flow, comparison between open channel flow and pipe flow, geometrical parameters of a channel, classification of open channels, classification of open channel flow, velocity distribution of channel section, uniform flow including continuity equation, energy equation, momentum equation, characteristics of uniform flow, Chezy's formula, Manning's formula, factors affecting Manning's roughness coefficient, most economical section of channel, computation of uniform flow and normal depth, non-uniform flow concepts such as specific energy, specific energy curve, critical flow, discharge curve, specific force, specific depth and critical depth, channel transitions, measurement of discharge and velocity using Venturi flume, standing wave flume, Parshall flume, broad crested weir, measurement of velocity by current meter, floats, hot-wire anemometer, gradually varied flow, dynamic equation of gradually varied flow, classification of channel bottom slopes and surface profiles, characteristics of surface profiles, computation of water surface profile by graphical, numerical and analytical methods including direct step, graphical integration and direct integration methods.	**16**

IV	Hydraulic Jumps, Flow through Pipes, and Computational Fluid Dynamics	Theory of hydraulic jump, elements and characteristics of hydraulic jump in a rectangular channel, length and height of jump, location of jump, types, applications and location of hydraulic jump, energy dissipation and other uses, surge as a moving hydraulic jump, positive and negative surges, dynamics of fluid flow including momentum principle and applications such as force on plates and pipe bends, moments of momentum equation, flow through pipes including loss of head through pipes, Darcy-Weisbach equation, minor losses, total energy equation, hydraulic gradient line, pipes in series, equivalent pipes, pipes in parallel, flow through laterals, flows in dead end pipes, siphon, power transmission through pipes, nozzles, analysis of pipe networks including Hardy Cross method, water hammer in pipes and control measures, branching of pipes, three reservoir problem, computational fluid dynamics basics including basic equations of fluid dynamics, grid generation, introduction to inviscid incompressible flow, boundary layer flow as applicable to CFD, hydroinformatics concepts and scope of internet and web-based modeling in water resources engineering.	
TOTAL			**66**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*22 NCH = 66 NCH	1*15 NCH = 15 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Hydraulic Engineering Lab Syllabus

Total Lab Hours for the semester = 30 (2 hours per week)		**Subject Code: CEE022C412**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Practical Work:

1. Flow Visualization
2. Studies in Wind Tunnel
3. Boundary Layer
4. Flow around an Aerofoil / circular cylinder
5. Uniform Flow
6. Velocity Distribution in Open channel flow
7. Venturi Flume
8. Standing Wave Flume
9. Gradually Varied Flow
10. Hydraulic Jump
11. Flow under Sluice Gate
12. Flow through pipes
13. Turbulent flow through pipes
14. Flow visualization
15. Laminar flow through pipes
16. Major losses / Minor losses in pipe

Text/Reference Books:

1. Fluid Mechanics, Sadhu Singh, Khanna Book Publishing Co., New Delhi
2. Fluid Machinery (Hydraulic Machines), Sadhu Sigh, Khanna Book Publishing Co.,
3. Hydraulics and Fluid Mechanics, P.M. Modi and S.M. Seth, Standard Book House
4. Theory and Applications of Fluid Mechanics, K. Subramanya, Tata McGraw Hill.
5. Open channel Flow, K. Subramanya, Tata McGraw Hill.
6. Open Channel Hydraulics, Ven Te Chow, Tata McGraw Hill.
7. Burnside, C.D., "*Electromagnetic Distance Measurement*," Beekman Publishers, 1971.

Subject Name: Transportation Engineering **Subject Code: CEE022C403**

L-T-P-C – 3-0-0-3 **Credit Units: 03** **Scheme of Evaluation: T**

Objective:

To understand the importance of transportation and characteristics of road transport and know about the history of highway development, surveys and classification of roads. The course is

designed to impart knowledge about the geometric design of highways, traffic characteristics and pavement materials

Prerequisites: Concepts of +2 level Physics

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** various engineering surveys for highways.	**BT 1**
CO 2	**Illustrate** tests on highway materials.	**BT 2**
CO 3	**Understand** geometric design and alignment concepts.	**BT 3**
CO 4	**Emphasize** the importance of traffic volume studies and traffic regulation.	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course Content**	**Periods**
I.	Highway development and planning	Highway development and planning- Classification of roads, road development in India, Current Road projects in India; highway alignment and project preparation.	**8**
	Geometric design of highways	Geometric design of highways-: Introduction; highway cross section elements; sight distance, design of horizontal alignment; design of vertical alignment; design of intersections, problems.	
II.	Accessibility to Differently Abled Publics.	Design of Access Routes &Walkways (Elements of walkways, Tactile Navigation Systems, BRT Systems, Pedestrian streets and other related aspects), Accessible Streets and Mobility Environments (Street Elements for Accessibility,	**7**

		dimensions and codes material, TGSIs), Inclusive Public Transportation System	
	Traffic engineering & control.	Traffic Characteristics, traffic engineering studies, traffic flow and capacity, traffic regulation and control; design of road intersections; design of parking facilities; highway lighting; problems	
III.	Pavement materials	Materials used in Highway Construction- Soils, Stone aggregates, bituminous binders, bituminous paving mixes; Portland cement and cement concrete: desirable properties, tests, requirements for different types of pavements. Problems	**8**
IV.	Design of pavements	Introduction; flexible pavements, factors affecting design and performance; stresses in flexible pavements; design of flexible pavements as per IRC; rigid pavements- components and functions; factors affecting design and performance of CC pavements; stresses in rigid pavements; design of concrete pavements as per IRC; problems	**7**
TOTAL			**30**

On completion of the course, the students will be able to:

- Carry out surveys involved in planning and highway alignment
- Design the geometric elements of highways and expressways
- Carry out traffic studies and implement traffic regulation and control measures and intersection design
- Characterize pavement materials and
- Design flexible and rigid pavements as per IRC

Transportation Engineering Lab Syllabus		**Subject Code: CEE022C413**
L-T-P-C – 0-0-2-1	**Credit Units:01**	**Scheme of Evaluation: P**

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

Objective: To determine the various parameters of road building materials and understand their physical properties.

1. Sieve analysis
2. Impact test
3. Crushing Strength test
4. Abrasion test
5. Water Absorption test
6. Specific gravity test
7. Flakiness Index test
8. Elongation Index test
9. CBR Test

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2 * 15 NCH = 30 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbooks:

1. Khanna, S.K., Justo, C.E.G and Veeraragavan, A, 'Highway Engineering', Revised 10th Edition, Nem Chand & Bros, 2017

2. Kadiyalai, L.R., 'Transportation Engineering', Khanna Book Publishing Co., New Delhi (ISBN: 978-9382-609-858)

3. Partha Chakraborty, ' Principles Of Transportation Engineering, PHI Learning,

4. Fred L. Mannering, Scott S. Washburn, Walter P. Kilareski,'Principles of Highway Engineering and Traffic Analysis', 4th Edition, John Wiley

5. Srinivasa Kumar, R, Textbook of Highway Engineering, Universities Press, 2011.

Subject Name: Surveying and Geomatics	**Subject Code: CEE022C404**	
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objective of this Course is to introduce students to the field of surveying as applied in civil engineering field. The course is also aimed at familiarizing students with various methods and instruments used for surveying.

Prerequisites: Concepts of +2 level Physics and Maths

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain the basics of surveying, including accurate measurements, field booking, and plotting.**	**BT 1**
CO 2	**Demonstrate proficiency in using survey instruments and tools.**	**BT 2**

CO 3	**Understand the relevance of geospatial information in civil engineering projects**	**BT 3**
CO 4	**Analyze survey data for decision-making and Explore different methods for data collection and processing in geomatics.**	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I.	***Introduction to Surveying***	**Principles, Linear, angular and graphical methods, Survey stations, Survey lines-ranging, Bearing of survey lines, Levelling: Plane table surveying, Principles of levelling-booking and reducing levels; differential, reciprocal leveling, profile levelling and cross sectioning. Digital and Auto Level, Errors in levelling; contouring: Characteristics, methods, uses; areas and volumes.**	**14**
	Triangulation and Trilateration	**Theodolite survey: Instruments, Measurement of horizontal and vertical angle; Horizontal and vertical control - methods -triangulation - network- Signals. Baseline - choices - instruments and accessories - extension of base lines - corrections - Satellite station - reduction to centre - Intervisibility of height and distances - Trigonometric leveling - Axis single corrections.**	
II.	***Curves***	**Elements of simple and compound curves – Method of setting out– Elements of Reverse curve - Transition curve – length of curve – Elements of transition curve - Vertical curves**	**14**

	Modern Field Survey Systems	**Principle of Electronic Distance Measurement, Modulation, Types of EDM instruments, Distomat, Total Station – Parts of a Total Station – Accessories –Advantages and Applications, Field Procedure for total station survey, Errors in Total Station Survey; Global Positioning Systems- Segments, GPS measurements, errors and biases, Surveying with GPS, Co-ordinate transformation, accuracy considerations.**	
III.	***Photogrammetry Surveying***	**Introduction, Basic concepts, perspective geometry of aerial photograph, relief and tilt displacements, terrestrial photogrammetry, flight planning; Stereoscopy, ground control extension for photographic mapping- aerial triangulation, radial triangulation, methods; photographic mapping- mapping using paper prints, mapping using stereoplotting instruments, mosaics, map substitutes.**	**8**
IV.	***Remote Sensing***	**Introduction –Electromagnetic Spectrum, interaction of electromagnetic radiation with the atmosphere and earth surface, remote sensing data acquisition: platforms and sensors; visual image interpretation; digital image processing.**	**9**
TOTAL			**45**

On completion of the course, the students will be able to:

- **Apply the knowledge, techniques, skills, and applicable tools of the discipline to engineering and surveying activities**
- **Translate the knowledge gained for the implementation of Civil infrastructure facilities**
- **Relate the knowledge on Surveying to the new frontiers of science like Hydrographic surveying, Electronic Distance Measurement, Global Positioning System, Photogrammetry and Remote Sensing.**

Surveying and Geomatics Lab Syllabus

Total Lab Hours for the semester = 30 (2 hours per week) **Subject Code: CEE022C414**

L-T-P-C – 0-0-2-1 **Credit Units:01** **Scheme of Evaluation: P**

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

Objective: To familiarize students with different surveying instruments and techniques like levelling, traversing, direct & indirect contouring etc.

1. **To Range a line more than one chain length and recording details in a field book.**
2. **Open and Closed Traversing using a Prismatic compass.**
3. **To determine the height of a building (accessible and inaccessible) using a theodolite.**
4. **Profile and Cross-section levelling using a dumpy level.**
5. **Repetition and Reiteration method of horizontal angle measurement by theodolite.**
6. **To locate a simple circular curve on field.**
7. **Plane table surveying**
8. **Indirect Contouring**
9. **Determination of area of an irregular boundary.**
10. **Total Station Demonstration/ Workshop.**

Credit Distribution

Lecture/ Tutorial	Practicum	Experiential Learning
3 * 15 NCH = 45 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbooks:

1. **Madhu, N, Sathikumar, R and Satheesh Gobi, Advanced Surveying: Total Station, GIS and Remote Sensing, Pearson India, 2006.**
2. **Manoj, K. Arora and Badjatia, Geomatics Engineering, Nem Chand & Bros, 2011**
3. **Bhavikatti, S.S., Surveying and Levelling, Vol. I and II, I.K. International, 2010**
4. **Garg, P.K., Principles and Theory of Geoinformatics, Khanna Publishing House, 2019.**
5. **Chandra, A.M., Higher Surveying, Third Edition, New Age International (P) Limited, 2002.**
6. **Anji Reddy, M., Remote sensing and Geographical information system, B.S. Publications, 2001.**
7. **Arora, K.R., Surveying, Vol-I, II and III, Standard Book House, 2015.**

Subject Name: Construction Engineering & Management		**Subject Code: CEE022C405**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to acquaint the students with various aspects of Construction Engineering & Management. They'll gain insights into construction processes, stakeholder roles (including architects, engineers, contractors, and clients), and modern construction practices. Additionally, they'll learn about project dynamics, effective planning, cost optimization, contract administration, and the importance of strong communication skills within the construction industry.

Prerequisites: NIL

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** key concepts related to construction practices, project development, and stakeholder roles.	**BT 1**
CO 2	**Explain** modern construction practices, including techniques, materials, and safety protocols. Also be able to explain the dynamics of construction projects, such as the interplay between stakeholders, project objectives, and resource allocation	**BT 2**
CO 3	**Apply** knowledge by planning, controlling, and monitoring construction projects with respect to both time and cost. Will be able to put theoretical concepts into practical use during project execution.	**BT 3**

CO 4	**Analyze** construction projects from an economic perspective, considering costs, benefits, and optimization strategies. Also critically evaluate contract structures and address issues related to	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course Content**	**Periods**
I.	**Basics of Construction** **Construction project planning**	Basics of Construction- Unique features of construction, construction projects- types and features, phases of a project, agencies involved and their methods of execution. Construction project planning- Stages of project planning: pre-tender planning, pre-construction planning, detailed construction planning, role of client and contractor, level of detail. Process of development of plans and schedules, work break-down structure, activity lists, assessment of work content, concept of productivities, estimating durations, sequence of activities, activity utility data; Techniques of planning- Bar charts, Gantt Charts. Networks: basic terminology, types of precedence relationships, preparation of CPM networks: activity on link and activity on node representation, computation of float values, critical and semi critical paths, calendaring networks. PERT- Assumptions underlying PERT analysis, determining	**12**

II.	**Construction Methods basics: Construction Equipment basics:**	Construction Methods basics: Types of foundations and construction methods; Basics of Formwork and Staging; Common building construction methods (conventional walls and slabs; conventional framed structure with blockwork walls; Modular construction methods for repetitive works; Precast concrete construction methods; Basics of Slip forming for tall structures; Basic construction methods for steel structures; Basics of construction methods for Bridges. Construction Equipment basics: Conventional construction methods Vs Mechanized methods and advantages of latter; Equipment for Earthmoving, Dewatering; Concrete mixing, transporting & placing; Cranes, Hoists and other equipment for lifting; Equipment for transportation of materials. Equipment Productivities	**12**

III.	**Planning and organizing construction site and resources- Project Monitoring & Control-**	Planning and organizing construction site and resources- Site: site layout including enabling structures, developing site organization, Documentation at site; Manpower: planning, organizing, staffing, motivation; Materials: concepts of planning, procurement and inventory control; Equipment: basic concepts of planning and organizing; Funds: cash flow, sources of funds; Histograms and S-Curves. Earned Value; Resource Scheduling- Bar chart, line of balance technique, resource constraints and conflicts; resource aggregation, allocation, smoothening and leveling. Common Good Practices in Construction. Project Monitoring & Control- Supervision, record keeping, periodic progress reports, periodical progress meetings. Updating of plans: purpose, frequency and methods of updating. Common causes of time and cost overruns and corrective measures. Basics of Modern Project management systems such as Lean Construction; Use of Building Information Modelling (BIM) in project management; Quality control: concept of quality, quality of constructed structure, use of manuals and checklists for quality control, role of inspection, basics of statistical quality control. Safety, Health and Environment on project sites: accidents; their causes, effects and preventive measures, costs of accidents, occupational health problems in construction, organizing for safety and health.	**11**

IV.	**Contracts Management basics: Construction Costs:**	Contracts Management basics: Importance of contracts; Types of Contracts, parties to a contract; Common contract clauses (Notice to proceed, rights and duties of various parties, notices to be given, Contract Duration and Price. Performance parameters; Delays, penalties and liquidated damages; Force Majeure, Suspension and Termination. Changes & variations, Dispute Resolution methods. Construction Costs: Make-up of construction costs; Classification of costs, time- cost trade-off in construction projects, compression and decompression.	**10**
TOTAL			**45**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH	NA	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

1. Sharma, S.C. & Deodhar S.V., Construction Engineering and Management, Khanna Book Publishing Co., 2022.
2. Varghese, P.C., "*Building Construction"*, Prentice Hall India, 2007.
3. *National Building Code,* Bureau of Indian Standards, New Delhi, 2017.

4. Chudley, R., *Construction Technology*, ELBS Publishers, 2007.

Reference Books:

1. Peurifoy, R.L. Construction Planning, Methods and Equipment, McGraw Hill, 2011
2. Nunnally, S.W. Construction Methods and Management, Prentice Hall, 2006
3. Rajoria, K.B., Case Studies in Construction Project Management, Khanna Publishing House, 2023.
4. Jha, Kumar Neeraj., Construction Project management, Theory & Practice, Pearson Education India, 2015
5. Punmia, B.C., Khandelwal, K.K., Project Planning with PERT and CPM, Laxmi

Subject Name: Geotechnical Engineering		**Subject Code: CEE022C406**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

 To introduce the fundamental concepts of soil mechanics, including soil formation, classification, and its physical and engineering properties relevant to civil engineering applications.

 To develop an understanding of the behavior of soils under various loading conditions, including concepts of permeability, seepage, stress distribution, and consolidation.

 To equip students with the ability to determine the shear strength of soils, enabling the analysis of slope stability, earth pressure, and bearing capacity problems.

 To provide hands-on experience in laboratory and field testing of soils, interpreting test results for practical engineering applications in foundation design and site investigation.

 To prepare students to analyze and design shallow and deep foundations, retaining structures, and other geotechnical systems with an understanding of soil-structure interaction.

 To foster awareness of modern tools and techniques in geotechnical investigations, including geophysical methods, ground improvement techniques, and sustainable geotechnical solutions.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**urse Outcome**	**Blooms Taxonomy**

		Level
CO 1	**Apply the fundamental principles of soil mechanics to classify soils and analyze their physical and engineering properties relevant to civil engineering projects.**	**BT 1**
CO 2	**Conduct laboratory and field tests to determine soil properties such as permeability, shear strength, compaction, and consolidation, and interpret the results for practical engineering applications.**	**BT 2**
CO 3	**Analyze stress distribution, seepage, and effective stress in soil masses, and assess their impact on soil stability and foundation performance.**	**BT 3**
CO 4	**Design safe and efficient shallow and deep foundation systems, retaining structures, and earthworks by integrating soil behavior analysis and modern geotechnical investigation techniques.**	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	Fundamentals of Soil Mechanics and Soil Properties	Introduction to types of soils, their formation and deposition, definitions of soil mechanics, soil engineering, rock mechanics, and geotechnical engineering, scope of soil engineering, comparison and difference between soil and rock, basic definitions and relationships including soil as a three-phase system in terms of weight, volume, voids ratio, and porosity, definitions of moisture content, unit weights, degree of saturation, voids ratio, porosity, specific gravity, mass specific gravity, relationships between volume weight, voids ratio and moisture content, unit weight and percent air voids, saturation and moisture content, moisture content and specific gravity, determination of various parameters such as moisture content by oven dry method, pycnometer, sand bath method, torsional balance method, nuclear method, alcohol method and sensors, specific gravity by density bottle method, pycnometer method, measuring flask method, unit weight by water displacement method, submerged weight method, core-cutter method, sand-replacement method.	**16**
II.	Soil Classification,	Plasticity characteristics of soil including introduction to definitions such as plasticity of soil, consistency limits—liquid limit, plastic limit, shrinkage limit—plasticity,	**17**

	Plasticity, and Permeability	liquidity and consistency indices, flow and toughness indices, definitions of activity and sensitivity, determination of liquid limit, plastic limit and shrinkage limit, use of consistency limits, classification of soils with particle size classification, textural classification, unified soil classification system, Indian standard soil classification system, field identification of soils and general characteristics of soils in different groups, permeability of soil covering Darcy's law, validity of Darcy's law, determination of coefficient of permeability by laboratory methods such as constant-head and falling-head methods, field methods including pumping-in and pumping-out tests, permeability of stratified soils, factors affecting permeability of soil, seepage analysis including introduction to stream and potential functions, characteristics of flow nets, and graphical methods to plot flow nets.	
III.	Soil Stresses, Compaction, and Consolidation	Effective stress principle including introduction, nature of effective stress, effect of water table, fluctuations of effective stress, effective stress in soils saturated by capillary action, seepage pressure, quick sand condition, compaction of soil covering introduction, theory of compaction, laboratory determination of optimum moisture content and maximum dry density, field compaction methods, compaction specifications and field control, stresses in soils including stresses due to point load, line load, strip load, uniformly loaded circular area, rectangular loaded area, influence factors, isobars, Boussinesq's equation, Newmark's influence chart, contact pressure under rigid and flexible area, computation of displacements from elastic theory, consolidation of soil including introduction, comparison between compaction and consolidation, initial, primary and secondary consolidation, spring analogy for primary consolidation, interpretation of consolidation test results, Terzaghi's theory of consolidation, final settlement of soil deposits, computation of consolidation settlement and secondary consolidation.	**16**
IV	Shear Strength, Slope Stability, and	Shear strength covering Mohr circle and its characteristics, principal planes, relation between major and minor principal stresses, Mohr-Coulomb theory, types of shear tests including direct shear test, merits of direct shear test, triaxial compression tests, behavior of UU, CU and CD tests, pore-pressure measurement,	

	Soil Exploration	computation of effective shear strength parameters, unconfined compression test, vane shear test, stability of slopes including introduction, types of slopes and their failure mechanisms, factor of safety, analysis of finite and infinite slopes, wedge failure, Swedish circle method, friction circle method, stability numbers and charts, soil exploration including introduction, methods of site exploration and soil investigation, methods of boring, soil samplers, sampling procedures, trial pits, borings, penetrometer tests, analysis of borehole logs, geophysical and advanced soil exploration methods.	
TOTAL			**66**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*22 NCH = 66 NCH	1*15 NCH = 15 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Subject Name: GEOTECHNICAL ENGINEERING LAB **Subject Code: CEE022C416**

L-T-P-C – 0-0-2-1 **Credit Units: 01** **Scheme of Evaluation: P**

Practical Work: List of tests on-

1. Field Density using Core Cutter method.
2. Field Density using Sand replacement method.
3. Natural moisture content using Oven Drying method.
4. Field identification of Fine-Grained soils.
5. Specific gravity of Soils.
6. Grain size distribution by Sieve Analysis.

7. Grain size distribution by Hydrometer Analysis.
8. Consistency limits by Liquid limit
9. Consistency limits by Plastic limit
10. Consistency limits by Shrinkage limit.
11. Permeability test using Constant-head test method.
12. Permeability test using Falling-head method.
13. Compaction test: Standard Proctor test.
14. Compaction test: Modified Proctor test.
15. Relative density.
16. Consolidation Test.
17. Triaxial Test (UU)
18. Vane shear test
19. Direct Shear Test

Unconfined Compression Strength Test.

Text/Reference Books:

1. Soil Mechanics by Craig R.F., Chapman & Hall
2. Elements of Land/ Soil Pollution by O.P. Gupta, Khanna Book Publishing.
3. Fundamentals of Soil Engineering by Taylor, John Wiley & Sons
4. An Introduction to Geotechnical Engineering, by Holtz R.D. and Kovacs, W.D., PrenticeHall, NJ
5. Principles of Geotechnical Engineering, by Braja M. Das, Cengage Learning
6. Principles of Foundation Engineering, by Braja M. Das, Cengage Learning
7. Essentials of Soil Mechanics and Foundations: Basic Geotechnics by David F. McCarthy
8. Soil Mechanics in Engineering Practice by Karl Terzaghi, Ralph B. Peck, andGholamreza Mesri.
9. Geotechnical Engineering: Principles and Practices of Soil Mechanics and Foundation Engineering (Civil and Environmental Engineering) by V.N.S. Murthy

SEMESTER –V

Subject Name: Design of RCC Structures		**Subject Code: CEE022C501**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objective of the Design of Reinforced Cement Concrete (RCC) subject is to equip students with the knowledge and skills required to design safe, durable, and economical concrete structures. It focuses on understanding the properties of concrete and steel, applying design philosophies such as limit state and working stress methods, and designing structural elements like beams, slabs, columns, and footings. The course emphasizes the use of IS codes (especially IS 456:2000), ensures structural safety and serviceability, and teaches proper reinforcement detailing. It also prepares students to tackle practical site-related challenges and promotes the use of sustainable and cost-effective design practices.

Prerequisites: Knowledge of Strength of Materials, Structural Analysis, Engineering Mechanics, Construction Materials, and Mathematics is essential for studying RCC Design.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**To understand** the concepts and assumptions behind Working Stress and Limit State Methods for RCC design.	**BT 1**
CO 2	**To apply** IS code provisions for the design of beams, slabs, columns, and footings using RCC design methods.	**BT 2**
CO 3	**To analyze** structural members for strength and serviceability under various loading conditions.	**BT 3**
CO 4	**To create** detailed design and reinforcement drawings of RCC and prestressed concrete components.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I.	**Working stress method and ultimate**	Reinforced Concrete Fundamentals (working Stress Method): Concept of reinforced concrete, stress strain characteristics of concrete and steel reinforcement, elastic theory, singly reinforced, balanced section, under reinforced section and over reinforced section.	**12**
II.	**Limit state method of design**	Concepts of probability and reliability, characteristic loads, characteristic strength, partial safety factors for loads and materials, limit states of collapse in flexure, direct compression, shear, limit states of serviceability in deflection and cracking, design of singly and doubly reinforced rectangular and T sections for flexure, design of members in shear and bond, design of columns for axial load, uni-axial bending moment, and bi-axial bending moment as per IS Code.	**12**
III.	**Limit state method of design**	Design of one-way and two-way slabs, design of beams subjected to bending and torsion, design of isolated square and rectangular footings subjected to axial load and moments, design of combined foundations, design of doglegged and open-well type staircases, design of flat slab, design of post-tensioned slab.	**11**

IV.	**Pre-stressed concrete**	Basic principles of pre-stressed concrete: materials used and their properties, methods and systems of pre-stressing. Losses in pre-stress, analysis of various types of sections subjected to pre-stress and external loads. Prestressed and post tensioned members, Different types of Prestressed sections, BoxSection, Girder	**10**
TOTAL			**45**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH	NA	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

1 Jain and Jaikrishna, Plain and Reinforced Concrete, Vol. I, Nemchand Brothers.(ISBN-8185240086/978-8185240084).

2 Shrikhant Vanakudre, Prestressed Concrete (Materials, Analysis and Design), Khanna Publishing House, (ISBN: 9789386173317)

3 V. L. Shah and Karve, Limit State Design - Reinforced Concrete Structures Publications. (ISBN-9788190371711/8190371711).

4 N. Krishna Raju, Pre-stressed Concrete, Tata McGraw Hill. (ISBN-9789387886209/9387886204).

Recommended Reading

1. P. Dayaratnam, Design of Reinforced Concrete Structures, Oxford & IBH. (ISBN-9789386479785/9386479788).

2. T.Y. Lin, Design of Prestressed Concrete Structures, John Wiley and Sons Inc., 2010. (ISBN-788126528035/978-8126528035).

3. P.D. Arthur and V. Ramkrishnan, Ultimate Strength Design for Structural Concrete, Wheeler&Co. Pvt Ltd. (ISBN- 0273403230, 978-0273403234).

4. B.P. Hughes, Limit State Theory for Reinforced Concrete Design, Pitman. (ISBN- 0273010239,978-0273010234).

5. IS456 (2000), Plain and Reinforced Concrete.

6. IS 875 (1987), Part I- Design Loads (Other than earthquake) for Buildings

and Structures (DeadLoads).
7. IS 875 (1987), Part II- Design Loads (Other than earthquake) for Buildings and Structures(Imposed Loads).
8. IS 875 (2015), Part III- Design Loads (Other than earthquake) for Buildings and Structures(Wind Loads).
9. IS 875 (1987), Part IV- Design Loads (Other than earthquake) for Buildings and Structures(Snow Loads).

Subject Name: Environmental Engineering		**Subject Code: CEE022C502**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objective of this course is to provide a comprehensive understanding of environmental engineering systems related to **water supply**, **wastewater**, **air**, **noise**, and **solid waste management**, along with associated infrastructure and regulatory frameworks. It covers sources and treatment of water, design of supply and sewerage systems, water quality standards, and wastewater treatment processes. The syllabus also includes air and noise pollution control, municipal and hazardous solid waste management, building plumbing systems, and the roles of government authorities in environmental regulation and infrastructure planning.

Prerequisites: Basic knowledge of chemistry, fluid mechanics, and environmental science is a prerequisite for studying Environmental Engineering.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	To **understand** the sources, quality parameters, and treatment methods of water and wastewater.	**BT 1**
CO 2	To **apply** design principles for water supply, sewerage, stormwater, and solid waste management systems.	**BT 2**
CO 3	**To analyze** the causes and impacts of air and noise pollution, and evaluate appropriate control measures.	**BT 3**
CO 4	**To design** integrated environmental engineering solutions considering plumbing, waste handling, and regulatory compliance.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I.	**Water Supply and Treatment**	Topics include sources of water, water quality issues, water quality requirements for beneficial uses, water quality standards, water quality indices, and water safety plans. It also covers water supply systems, the need for planned water supply schemes, and water demand for domestic, industrial, and agricultural purposes. Additional topics include components of water supply systems, transmission of water, distribution systems, various valves used in water supply, service reservoirs and their design. Treatment processes include aeration, sedimentation, coagulation and flocculation, filtration, disinfection, and advanced treatment methods such as adsorption, ion exchange, and membrane processes.	**12**
II.	**Wastewater Engineering**	This module includes domestic and storm water, quantity of sewage, sewage flow variations, types and shapes of sewers, design parameters, operation and maintenance of sewers, sewage pumping, and sewerage systems. It also covers sewer appurtenances, design of sewerage systems, small bore systems, storm water quantification and design, sewage and sullage, and pollution due to improper sewage disposal. Further topics include National River Cleaning Plans, aerobic and anaerobic treatment systems, suspended and attached growth systems, sewage recycling, and quality requirements for various reuse purposes.	**12**

III.	**Air and Noise Pollution**	Topics covered include the composition and properties of air, quantification and monitoring of air pollutants, occupational and urban air pollution, automobile pollution, chemistry of combustion, automobile engines, fuel quality, and their operating conditions and interrelations. The module also addresses air quality standards, control measures for air pollution, and their limitations. It further includes basic concepts of noise pollution, noise measurement techniques, and noise control methods.	**11**
IV.	**Solid Waste Management and Building Plumbing**	This module covers municipal solid waste (MSW), its composition, chemical and physical parameters, and management processes including collection, transport, treatment, and disposal. It also includes special MSW from commercial establishments, construction sites, and biomedical sources. Topics on environmental effects of solid waste, health hazards, segregation, source reduction, recovery, recycling, and integrated solid waste management are also addressed. Hazardous waste types and classifications are included. The module concludes with building plumbing systems, plumbing in high-rise buildings, pressure reducing valves, break pressure tanks, storage tanks, building drainage, fixtures and fittings, and the role of government authorities in water supply, sewerage disposal, and environmental pollution control.	**10**
TOTAL			**45**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH	NA	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Subject Name: Environmental Engineering Lab	Subject Code: CEE022C512	
L-T-P-C – 0-0-2-1	Credit Units: 01	Scheme of Evaluation: P

Practical Work: List of Experiments

1. Physical Characterization of water: Turbidity, Electrical Conductivity, pH
2. Analysis of solids content of water: Dissolved, Settleable, suspended, total, volatile,inorganic etc.
3. Alkalinity and acidity, Hardness: total hardness, calcium and magnesium hardness
4. Analysis of ions: copper, chloride and sulfate
5. Optimum coagulant dose
6. Chemical Oxygen Demand (COD)
7. Dissolved Oxygen (D.O) and Biochemical Oxygen Demand (BOD)
8. Break point Chlorination
9. Bacteriological quality measurement: MPN,
10. Ambient Air quality monitoring (TSP, RSPM, SOx, NOx)
11. Ambient noise measurement

Text/Reference Books:

1. Environmental Engineering, M.P. Poonia, SC. Sharma, Santosh Kumar, Khanna BookPublishing Co., New Delhi.
2. Air Pollution and Control, Keshav Kant, Rajni Kant, Khanna Book Publishing Co.,New Delhi.
3. Introduction to Environmental Engineering and Science by Gilbert Masters, PrenticeHall, New Jersey.
4. Introduction to Environmental Engineering by P. Aarne Vesilind, Susan M. Morgan,Thompson /Brooks/Cole; Second Edition 2008.
5. Peavy, H.s, Rowe, D.R, Tchobanoglous, G. *Environmental Engineering*, Mc-Graw -Hill International Editions, New York 1985.
6. MetCalf and Eddy. *Wastewater Engineering, Treatment, Disposal and Reuse*, TataMcGraw-Hill, New Delhi.
7. Manual on Water Supply and Treatment. Ministry of Urban Development, New Delhi.
8. Plumbing Engineering. Theory, Design and Practice, S.M. Patil, 1999
9. Gupta, O.P., Elements of Water Pollution Control Engineering, Khanna PublishingHouse, New Delhi (ISBN: 9789386173225)
10. Gupta, O.P., Elements of Land/ Soil Pollution, Khanna Publishing House, New Delhi(ISBN: 9789382609735)

11. Gupta, O.P., Elements of Environmental Pollution Control, Khanna Publishing House,New Delhi (ISBN: 9789382609667)
12. Gupta, O.P., Khanna's Objective Types Questions & Answers in Environmental Engineering, Khanna Book Publishing Co.

Subject Name: Engineering Economics, Estimation & Costing
Subject Code: CEE022C503
L-T-P-C – 3-0-0-3 **Credit Units: 03** **Scheme of Evaluation: T**

Objective:

The objective of this course is to provide students with a comprehensive understanding of engineering economics with a focus on its application in infrastructure and construction projects. It aims to equip learners with the ability to analyze and compare economic alternatives using techniques such as present worth, future worth, annual worth, benefit-cost ratio, life cycle costing, and breakeven analysis. The course also emphasizes understanding technical specifications, cost estimation, and rate analysis for various construction components. Additionally, it introduces students to the concepts of competitive bidding and tender preparation, enabling them to evaluate project costs effectively and prepare economically sound proposals.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	To understand fundamental economic principles and their application in the Indian context, particularly for public and private sector projects.	**BT 1**
CO 2	To apply techniques such as present worth, future worth, annual worth, benefit-cost ratio, life cycle cost, and breakeven analysis for evaluating project alternatives.	**BT 2**
CO 3	**To analyze** technical specifications and quantify the cost components of a structure through material estimation and rate analysis.	**BT 3**
CO 4	**To evaluate** tendering procedures and prepare competitive bidding proposals for construction projects.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I.	**Fundamentals of Engineering Economics**	Introduction to economics, importance of economics in engineering, economics of India, comparison between the public and private sectors, role of economics in infrastructure and construction projects.	**12**
II.	**Time Value of Money and Financial Analysis**	Perform and evaluate present worth, future worth, and annual worth analyses on one or more economic alternatives, carry out and evaluate benefit/cost analysis, life cycle cost analysis, breakeven analysis on one or more economic alternatives.	**12**
III.	**Cost Evaluation and Technical Specifications**	Understand the technical specifications for various works to be performed for a project, analyze how technical specifications impact the cost of a structure, quantify the worth of a structure by evaluating quantities of constituents, derive their cost rates, build up the overall cost of the structure.	**11**
IV.	**Estimation, Bidding, and Project Costing**	Understand how competitive bidding works, prepare and submit a competitive bid proposal, comprehend procedures of tendering and contracting, apply cost estimation techniques in real-world construction projects.	**10**
TOTAL			**45**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH	NA	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Subject Name: Hydrology and Water Resource Engineering
Subject Code: CEE022C504

L-T-P-C – 3-0-0-3 **Credit Units: 03** **Scheme of Evaluation: T**

Objective:

The objective of this course is to provide students with a fundamental understanding of the hydrologic cycle and its components, including precipitation, evaporation, infiltration, runoff, and groundwater flow. It aims to equip students with analytical and design skills for planning and managing water resources systems such as reservoirs, canals, and flood control structures. The course also focuses on the application of hydrological models, data analysis techniques, and design methods in solving real-world problems related to water availability, distribution, and conservation.

Prerequisites: Basic knowledge of fluid mechanics and engineering mathematics.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**To understand** the components of the hydrologic cycle and the fundamental processes such as precipitation, evaporation, infiltration, runoff, and groundwater flow.	**BT 1**
CO 2	**To apply** analytical methods and hydrologic techniques for estimation of surface and groundwater flow, irrigation requirements, and reservoir design.	**BT 2**
CO 3	**To analyze** rainfall-runoff relationships, hydrographs, and well hydraulics using standard methods such as SCS-CN, unit hydrograph, and aquifer tests.	**BT 3**
CO 4	**To evaluate** the design and performance of water resources structures including canals, dams, spillways, and irrigation systems for efficient water distribution and flood control.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I.	**Introduction and Precipitation**	hydrologic cycle, water-budget equation, history of hydrology, worldwater balance, applications in engineering, sources of data., forms of precipitation, characteristics of precipitation in India, measurement of precipitation, rain gauge network, mean precipitation over an area, depth- area-duration relationships, maximum intensity/depth-duration-frequency relationship, Probable Maximum Precipitation (PMP), rainfall data in India.	**12**
II.	**Abstractions from precipitation and runoff**	evaporation process, evaporimeters, analytical methods of evaporation estimation, reservoir evaporation and methods for its reduction, evapotranspiration, measurement of evapotranspiration, evapotranspiration equations, potential evapotranspiration over India, actual evapotranspiration, interception, depression storage, infiltration, infiltration capacity, measurement of infiltration, modelling infiltration capacity, classification of infiltration capacities, infiltration indices. runoff volume, SCS-CN method of estimating runoff volume, flow- duration curve, flow-mass curve, hydrograph, factors affecting runoff hydrograph, componentsof hydrograph, base flow separation, effective rainfall, unit hydrograph surface water resourcesof India, environmental flows.	**12**
III.	**Ground water and well hydrology and Water withdrawals and uses**	forms of subsurface water, saturated formation, aquifer properties, geologic formations of aquifers, well hydraulics: steady state flow in wells, equilibrium equations for confined and unconfined aquifers, aquifer tests and water for energy production, water for agriculture, water for	**11**

		hydroelectric generation; flood control. Analysis of surface water supply, Water requirement of crops-Crops and crop seasons in India, cropping pattern, duty and delta; Qualityof irrigation water; Soil-water relationships, root zone soil water, infiltration, consumptive use, irrigation requirement, frequency of irrigation; Methods of applying water to the fields: surface,sub-surface, sprinkler and trickle / drip irrigation.	
IV.	**Distribution systems and Dams and spillways**	canal systems, alignment of canals, canal losses, estimation of design discharge. Design of channels- rigid boundary channels, alluvial channels, Kennedy'sand Lacey's theory of regime channels. Canal outlets: non-modular, semi-modular and modular outlets. Water logging: causes, effects and remedial measures. Lining of canals, types of lining.Drainage of irrigated lands: necessity, methods and embankment dams: Classification, design considerations, estimation and control of seepage, slope protection. Gravity dams: forces on gravity dams, causes of failure, stress analysis, elementary and practical profile. Arch and buttress dams. Spillways: components of spillways, types of gates for spillway crests; Reservoirs- Types, capacity of reservoirs, yield of reservoir, reservoir regulation, sedimentation, economic height of dam, selection of suitable site.	**10**
TOTAL			**45**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH	NA	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text/Reference Books:

1. K Subramanya, Engineering Hydrology, Mc-Graw Hill.O.P. Gupta, Elements of Water Pollution Control Engineering, Khanna Book Publishing Co.
2. O.P. Gupta, Multi-Choice Environmental Engineering, Khanna Book Publishing Co.
3. K N Muthreja, Applied Hydrology, Tata Mc-Graw Hill.
4. K Subramanya, Water Resources Engineering through Objective Questions, Tata Mc-Graw Hill.
5. Santosh Kumar, Irrigation Engineering, Khanna Book Publishing Co.
6. G L Asawa, Irrigation Engineering, Wiley Eastern
7. L W Mays, Water Resources Engineering, Wiley.
8. J D Zimmerman, Irrigation, John Wiley & Sons
9. C S P Ojha, R Berndtsson and P Bhunya, Engineering Hydrology, Oxford.

Subject Name: Hydrology lab **Subject Code: CEE022C514**

L-T-P-C – 0-0-2-1 **Credit Units: 01** **Scheme of Evaluation: T**

LIST OF EXPERIMENTS

Exp. No.	Name of Experiment	Relevant Standard / Reference
1	Determination of Gravimetric Moisture Content of Soil using Hot Air Oven and Electronic Balance	IS 2720 (Part 2): 1973 (Reaffirmed 2020) – Determination of Water Content
2	Determination of Soil Bulk Density, Dry Density and Porosity using Core Cutter Method	IS 2720 (Part 29): 1975 – Determination of Dry Density by Core Cutter Method
3	Determination of Soil Infiltration Rate using Double Ring Infiltrometer and Development of Infiltration Curve (Horton's Equation)	IS 15792: 2008 – Construction of Double Ring Infiltrometer; ASTM D3385/D3385M – Standard Test Method for Infiltration Rate of Soils
4	Determination of Field Capacity and Permanent Wilting Point of Soil	USDA Soil Survey Manual; FAO Irrigation and Drainage Paper No. 56
5	Estimation of Crop Water Requirement and Evapotranspiration	FAO Irrigation and Drainage Paper No. 56 – Crop Evapotranspiration (Penman–Monteith Method)

6	Estimation of Runoff using Rational Method and SCS Curve Number Method	USDA-NRCS National Engineering Handbook, Part 630 (Hydrology); IS 12056 (Hydrologic Design Practices) *(or standard hydrology references)*
7	Determination of pH, Electrical Conductivity and Refractive Index of Irrigation Water	IS 3025 (Part 11): pH; IS 3025 (Part 14): Electrical Conductivity; APHA Standard Methods for Examination of Water and Wastewater (Latest Edition)
8	Assessment of Irrigation Water Suitability based on Salinity Hazard, Sodium Hazard (SAR) and Irrigation Water Quality Criteria	FAO Irrigation and Drainage Paper No. 29 Rev.1 – Water Quality for Agriculture; US Salinity Laboratory (USSL) Handbook 60
9	Measurement and Analysis of Soil Temperature at Different Depths using Soil Thermometer	WMO Guide to Hydrological Practices (Latest Edition); FAO Soil Measurement Guidelines
10	GIS-Based Watershed Delineation and Drainage Analysis	IS 15792, NRSC–ISRO Watershed Atlas Guidelines, USGS/ArcHydro Technical Documentation, BIS Watershed Management Guidelines

SEMESTER –VI

Subject Name: Design of Steel Structures		Subject Code: CEE022C601
L-T-P-C – 4-0-0-4	Credit Units: 04	Scheme of Evaluation: T

Objective:

To impart knowledge and skills related to the design and detailing of structural steel connections, roofing systems, flooring systems, and columns with bases. The course emphasizes understanding load effects, structural behavior, and code-based design procedures for steel members and connections used in building construction.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Design and analyze bolted, welded, and riveted connections under axial and eccentric loads.**	**BT 1**
CO 2	**Evaluate and design roofing systems considering dead, imposed, and wind loads with proper detailing of trusses and their connections.**	**BT 2**
CO 3	**Develop appropriate floor systems using rolled steel sections, including the design of simply supported main and secondary beams.**	**BT 3**
CO 4	**Design steel columns and base plates, including detailing of lacing, battens, and column bases under axial and combined loading.**	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Bolted and welded connections**	Introduction to riveted connection, design of bolted and welded connections, axially and eccentrically loaded joints, simple connection of bracket plates to columns, beam to beam and beam to column connections, design of framed, unstiffened and stiffened seat connections.	**16**

II.	**Roofing system**	Imposed loads on flat and sloping roofs and floors, wind loads on sloping roofs and vertical cladding including effect of degree permeability and wind drag, analysis of pin-jointed trusses under various loading cases, computation of design forces in members, design and detailing of connections and supports.	**17**
III.	**Flooring system**	Concept of floor system with secondary beams, main beams and columns, design of simply supported beams using rolled steel sections.	**16**
IV	**Columns and bases**	Design of columns under axial loads using single or multiple rolled steel sections, design of lacing, battens, columns subjected to axial load and bending, design of slab and Gusseted base.	
TOTAL			**66**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*22 NCH = 66 NCH	1*15 NCH = 15 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. S.K. Duggal, Limit State Design of Steel Structures, Tata McGraw Hill Education Private Limited,2017. (ISBN: 9789351343493/9351343499).
2. V.L. Shah and V. Gore, Limit State Design of Steel Structures IS:800-2007, StructuresPublication, 2012. (ISBN: 90371754).

Recommended Reading:

1. S.S. Bhavikatti, Design of Steel Structures, I.K. International Publishing House Limited,2017.(ISBN: 9789385909559/938590955X).
2. N. Subramanian, Design of Steel Structures, Oxford University Press, 2011. (ISBN:9780198068815/0198068816).
3. IS 800 (2007), General Construction in Steel- Code of Practice, Ced 7: Structural Engineering andStructural Section, Published by Bureau of Indian Standard ManakBhavan, New Delhi.
4. IS 875- Part 1 (1987): Dead Loads, Indian Standard Code of Practice for Design Loads (Other thanEarthquake) for Buildings and Structures, Published by Bureau of Indian Standard
   Manak Bhavan, New Delhi.

IS 875- Part 2 (1987): Imposed Loads, Indian Standard Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures, Published by Bureau of Indian Standard Manak Bhavan, New Delhi.

5. IS 875- Part 3 (2017): Wind Loads, Indian Standard Code of Practice for Design Loads (Otherthan Earthquake) for Buildings and Structures, Published by Bureau of Indian Standard
   Manak Bhavan, New Delhi.

6. IS 875- Part 4 (1987): Snow Loads, Indian Standard Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures, Published by Bureau of Indian Standard Manak Bhavan, New Delhi.

7. IS 875- Part 5 (1987): Special Loads and Combinations, Indian Standard Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures, Published by Bureau ofIndian Standard Manak Bhavan, New Delhi

Subject Name: Sustainable and Green Construction	Subject Code: CEE022C603	
L-T-P-C – 3-0-0-3	Credit Units: 03	Scheme of Evaluation: T

Objective:

To provide students with a comprehensive understanding of sustainable development principles and their application in construction practices. The course aims to equip learners with knowledge of environmental systems, sustainable building materials, energy and water efficiency, green certifications, policy frameworks, and the ethical and economic aspects of sustainable construction.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain the principles of sustainable development and assess their relevance in the context of construction.**	**BT 1**
CO 2	**Analyze and apply sustainable materials, energy-efficient technologies, and water management strategies in construction projects.**	**BT 2**
CO 3	**Evaluate green building standards, certifications, and environmental policies relevant to sustainable construction practices.**	**BT 3**
CO 4	**Integrate ethical, economic, and project management strategies to promote sustainability in construction and infrastructure development.**	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Introduction to sustainable Development, Environment Science and Ecology,**	Definition and principles of sustainable development, Historical context and evolution of sustainability, Global and local challenges related to sustainability. Fundamentals of ecology,	**16**

	Sustainable construction Material and Technology:	Biodiversity and conservation, Climate change and its impacts, Environmental pollution and waste management. Green building materials and technologies,Life cycle assessment of materials, Resource efficiency and waste reduction in construction, Low-impact construction methods	
II.	**Energy Efficiency in Buildings, Water Management in Construction, Sustainable Site Planning**	Principles of energy-efficient design, Renewable energy integration in building design, Energy-efficient HVAC systems, Building energy modeling and simulation Sustainable water use in construction processes, Rainwater harvesting and greywater recycling, Water-efficient construction practices, Strategies for mitigating water pollution on construction sites Site selection and evaluation for sustainable construction, green space planning and landscaping, Sustainable stormwater management,	**17**
III.	**Sustainable Building Design, Construction and Demolition Waste Management, Building Certifications and Standards, Environmental Regulations and Policy**	Principles of green building design, Passive design strategies for energy efficiency, Daylighting and natural ventilation, Integration of sustainable technologies in design. Waste reduction and recycling in construction, Responsible demolition practices,Circular economy concepts in construction. LEED (Leadership in Energy and Environmental Design) certification, BREEAM (Building Research Establishment Environmental Assessment Method), Other regional and international green building certifications, Building codes and regulations for sustainable construction - Environmental impact assessments - Government policies promoting green construction	**16**

IV	**Economics of Sustainable Construction, Project Management for Sustainable Construction, Ethics and Social Responsibility in Construction**	Cost-benefit analysis for green building projects - Financial incentives for sustainable construction - Business models for green technologies in construction Sustainable project planning and execution - Stakeholder engagement and communication - Monitoring and evaluating sustainability performance, Ethical considerations in sustainable construction - Social impacts of green construction projects - Community engagement andstakeholder involvement	
	TOTAL		**66**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*22 NCH = 66 NCH	1*15 NCH = 15 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Textbooks / Recommended Readings:

1. ***Charles J. Kibert – Sustainable Construction: Green Building Design and Delivery, Wiley, Latest Edition.***
2. ***Edward A. McBean and Anjali A. Lalani – Sustainability and Sustainable Development, CRC Press.***
3. ***Barbara A. Richardson – Sustainable Building Systems and Construction for Designers, Fairchild Books.***
4. ***Richard Heinberg & David Fridley – Our Renewable Future: Laying the Path for One Hundred Percent Clean Energy, Island Press.***
5. ***Mehta, P. Kumar, and Monteiro, Paulo J.M. – Concrete: Microstructure, Properties, and Materials, McGraw Hill Education.***
6. ***Gottfried, David – Greed to Green: How Cities, Businesses, and Homes Can Save the Planet, Gibbs Smith.***
7. ***Yudelson, Jerry – Green Building A to Z: Understanding the Language of Green Building, New Society Publishers.***
8. ***M. Asif & T. Muneer – Energy Supply, Its Demand and Security Issues, Elsevier.***

Subject Name: Structural Analysis II	**Subject Code: CEE022C604**	
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Course Objective

The objective of this course is to enable students to analyze indeterminate structures using classical and matrix methods, evaluate the behavior of structures under moving and dynamic loads, and develop analytical skills required for the design of complex structural systems.

Course Outcomes (COs)

Upon successful completion of the course, students will be able to:

CO	Course Outcome	Bloom's Level
CO1	Explain the concepts of indeterminate structures and evaluate forces and displacements using energy methods.	BT2
CO2	Analyze continuous beams and rigid frames using slope-deflection and moment distribution methods.	BT3
CO3	Analyze indeterminate structures using matrix methods and evaluate structural stiffness characteristics.	BT4
CO4	Evaluate the behavior of structures subjected to moving and dynamic loads and interpret analytical results for engineering applications.	BT5

Detailed Syllabus

Module	Topics	Course Contents	Hours
I	**Energy Methods and Analysis of Indeterminate Structures**	Review of statically indeterminate structures; strain energy; Castigliano's First and Second Theorems; unit load method; Maxwell's reciprocal theorem; calculation of slopes and deflections of beams, trusses and frames; temperature effects and support settlements.	**10**
II	**Classical Methods of Structural Analysis**	Slope-deflection method; member stiffness; fixed-end moments; analysis of continuous beams; analysis of rigid frames with and without side sway; Moment	**12**

Module	Topics	Course Contents	Hours
		Distribution Method; distribution and carry-over factors; application to continuous beams and frames.	
III	**Matrix Methods of Structural Analysis**	Introduction to matrix methods; flexibility and stiffness concepts; coordinate systems; element stiffness matrix; global stiffness matrix; transformation matrix; analysis of trusses, beams and plane frames using stiffness method; introduction to computer applications in structural analysis.	**12**
IV	**Moving Loads and Introduction to Dynamic Analysis**	Influence lines for indeterminate structures; moving loads on beams and bridges; envelopes of forces; fundamentals of structural dynamics; degrees of freedom; free and forced vibration of single-degree-of-freedom systems; damping; resonance; introduction to earthquake response of structures.	**11**

Total Hours: 45

Text Books

1. C.S. Reddy, **Basic Structural Analysis**, 4th Edition, Tata McGraw-Hill.
2. R.C. Hibbeler, **Structural Analysis**, 10th Edition, Pearson.
3. Devdas Menon, **Advanced Structural Analysis**, Narosa Publishing.
4. S.S. Bhavikatti, **Structural Analysis Vol. II**, Vikas Publishing.

Reference Books

1. A.K. Jain, **Advanced Structural Analysis**, Nem Chand & Bros.
2. Wang, Salmon and Pincheira, **Matrix Structural Analysis**, Wiley.
3. Aslam Kassimali, **Structural Analysis**, Cengage Learning.
4. T.S. Thandavamoorthy, **Structural Analysis**, Oxford University Press.
5. Weaver & Gere, **Matrix Analysis of Framed Structures**, CBS Publishers.

SEMESTER –VII

Subject Name: Foundation engineering	**Subject Code: CEE022C70X**
L-T-P-C –4 -0-0-4	**Credit Units: 04** **Scheme of Evaluation: T**

Prerequisites:

- Basic knowledge of Soil Mechanics.
- Understanding of Engineering Geology and Surveying.
- Fundamentals of Strength of Materials and Structural Analysis.
- Basic knowledge of Reinforced Concrete Design.

Module	**Learning Objectives**	**Teaching Learning Process**	**Learning Outcomes**	**Course Evaluation**
I. Introduction to Foundation Engineering & Soil Exploration	Help students understand the importance of foundations, site investigation and subsurface exploration.	1. Lectures 2. Assignments 3. PowerPoint Presentations 4. NPTEL Videos 5. Textbooks	Students will understand the selection of suitable foundations based on soil conditions and loading.	**Semester End Examination:** 70 Marks **Internal Assessment:** 30 Marks *(Assignment: 05, Quiz: 05, Presentation: 05, Mid-Term: 10, Attendance: 05)*
II. Shallow Foundations	Enable students to understand bearing capacity, settlement and design of shallow foundations.	Same as above	Students will analyse bearing capacity and design isolated, strip and combined footings.	

Module	Learning Objectives	Teaching Learning Process	Learning Outcomes	Course Evaluation
III. Raft and Deep Foundations	Introduce raft foundations, pile foundations and well foundations.	Same as above	Students will determine pile capacity, pile group behaviour and select suitable deep foundations.	
IV. Special Foundations & Retaining Structures	Develop understanding of retaining walls, machine foundations and ground improvement techniques.	Same as above	Students will evaluate special foundation systems and recommend suitable solutions for difficult ground conditions.	

Detailed Syllabus

Modules	Topics / Course Content	Hrs.	Marks
I	**Introduction to Foundation Engineering:** Functions and requirements of foundations; types of foundations; factors affecting foundation selection; site investigation; methods of soil exploration; boring and sampling techniques; Standard Penetration Test (SPT); Plate Load Test; interpretation of soil investigation reports; groundwater table; Safe Bearing Capacity (SBC); Allowable Bearing Pressure; relevant IS Codes.	**15**	**25**
II	**Shallow Foundations:** Bearing capacity theories; Terzaghi's bearing capacity theory; effect of water table; eccentric and inclined loading; settlement of foundations; immediate and consolidation settlement;	**10**	**25**

Modules	Topics / Course Content	Hrs.	Marks
	design principles of isolated footing, strip footing, wall footing and combined footing; introduction to raft foundations.		
III	**Raft and Deep Foundations:** Types and applications of raft foundations; contact pressure; compensated foundations; classification of piles; load carrying capacity of piles; static and dynamic pile formulae; pile group efficiency; pile load tests; under-reamed piles; well foundations and caissons; construction methods.	**10**	**25**
IV	**Special Foundations & Modern Developments:** Machine foundations; vibration concepts; retaining walls; Rankine's and Coulomb's earth pressure theories; sheet piles; foundations on expansive soils and black cotton soils; ground improvement techniques; geosynthetics; stone columns; soil nailing; micropiles; sustainable and innovative foundation technologies.	**10**	**25**
	Total	**45**	**100**

Textbooks

- Das, B. M. & Sobhan, K. **Principles of Foundation Engineering**, Cengage Learning.
- Punmia, B. C., Jain, A. K. & Jain, A. K. **Soil Mechanics and Foundations**, Laxmi Publications.
- Arora, K. R. **Soil Mechanics and Foundation Engineering**, Standard Publishers.
- Varghese, P. C. **Foundation Engineering**, PHI Learning.

Reference Books

1. Bowles, J. E. *Foundation Analysis and Design*, McGraw-Hill.
2. Coduto, D. P., Yeung, M. C. R. & Kitch, W. A. *Foundation Design: Principles and Practices*, Pearson.

3. Gopal Ranjan & Rao, A. S. R. *Basic and Applied Soil Mechanics*, New Age International.

4. Tomlinson, M. J. & Woodward, J. *Pile Design and Construction Practice*, CRC Press.

5. IS 6403 – Determination of Bearing Capacity of Shallow Foundations.

6. IS 1904 – Design and Construction of Foundations.

7. IS 2911 (Parts 1–4) – Design and Construction of Pile Foundations.

8. IS 8009 – Settlement of Foundations.